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Lamp constructs comprising allergens

Abstract

The present invention provides improved LAMP Constructs comprising specific fragments of the LAMP luminal domain to deliver allergens to immune cells for enhanced processing. These LAMP Constructs can be used for the treatment of disease and in particular allergic reactions and/or allergies. The improved LAMP Constructs allow for presentation of properly configured three dimensional epitopes for production of an immune response when administered to a subject. The improved LAMP Constructs can be multivalent molecules, and/or can be provided as part of a multivalent vaccine containing two or more LAMP Constructs.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11203629	12/2020	Heiland	N/A	A61P 35/00
11773153	12/2022	Heiland	N/A	C07K 14/70596
2007/0269457	12/2006	Niazi et al.	N/A	N/A
2012/0294879	12/2011	August et al.	N/A	N/A
2016/0185831	12/2015	Hearl et al.	N/A	N/A
2016/0271245	12/2015	Hearl et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2004537285	12/2003	JP	N/A
2012500829	12/2011	JP	N/A
2013503159	12/2012	JP	N/A
2015521467	12/2014	JP	N/A
2019537447	12/2018	JP	N/A
2020517271	12/2019	JP	N/A
2020518613	12/2019	JP	N/A
2002080851	12/2001	WO	N/A
2010023247	12/2009	WO	N/A
2011031298	12/2010	WO	N/A
2011046996	12/2010	WO	N/A
2013187906	12/2012	WO	N/A
2015200357	12/2014	WO	N/A
2017020026	12/2016	WO	N/A

2018093932	12/2017	WO	N/A
2018195527	12/2017	WO	N/A
2018204534	12/2017	WO	N/A

OTHER PUBLICATIONS

Preliminary Report on Patentability dated Nov. 26, 2020 and received in PCT/US2019/032305. cited by applicant

Carlsson et al., "Assignment of O-Glycan Attachment Sites to the Hinge-like Regions of Human Lysosomal Membrane Glycoproteins Lamp-1 and Lamp-2," *Archive of Biochemistry and Biophysics*, 1993, 304(1): 65-73. cited by applicant

Godinho et al., "Regulation of HIV-Gag Expression and Targeting to the Endolysosomal/Secretory Pathway by the Luminal Domain of Lysosomal-Associated Membrane Protein (LAMP-1) Enhance Gag-Specific Immune Response," *PLOS One*, 2014, 9(6):e99887, 11 pages. cited by applicant

Su et al., "CryJ-LAMP DNA Vaccines for Japanese Red Cedar Allergy Induce Robust Th1-Type Immune Responses in Murine Model," *J Immunol Res*, 2016, Article ID 4857869, 15 pages. cited by applicant

Wilke et al., "Crystal structure of the conserved domain of the DC lysosomal associated membrane protein: implications for the lysosomal glycocalyx," *BMC Biology*, 2012, 10:62, 15 pages. cited by applicant

International Search Report, Written Opinion and Search Strategy dated Sep. 2, 2019 and received in PCT/US2019/032305. cited by applicant

Tan, L. et al., "Intramuscular Immunization with DNA Construct Containing Der p 2 and Signal Peptide Sequences Primed Strong IgE Production", *Vaccine*, vol. 24, pp. 5762-5771, (2006). cited by applicant

Arruda et al., "Dendritic Cell-Lysosomal-Associated Membrane Protein (LAMP) and LAMP-1-HIV-1 Gag Chimeras Have Distinct Cellular Trafficking Pathways and Prime T and B Cell Responses to a Diverse Repertoire of Epitopes," *J Immunology*, 2006, 177: 2265-2275. cited by applicant

Arterburn et al., "The Disulfide Structure of Mouse Lysosome-associated Membrane Protein 1," *J. Biol. Chem.*, 1990, 265:7419-7423. cited by applicant

Carlsson et al., "Structure of Human Lysosomal Membrane Glycoprotein 1, Assignment of Disulfide Bonds and Visualization of its Domain Arrangement," *J. Biol. Chem.*, 1989, 264(34):20526-20531. cited by applicant

De Arruda et al. "DNA vaccine encoding human immunodeficiency virus-1 Gag, targeted to the major histocompatibility complex II compartment by lysosomal-associated membrane protein, elicits enhanced long-term memory response," *Immunology*, 2004, 112(1):126-33. cited by applicant

Godinho et al., "Regulation of HIV-Gag Expression and Targeting to the Endolysosomal /Secretory Pathway by the Luminal Domain of Lysosomal-Associated Membrane Protein (LAMP-1) Enhance Gag-Specific Immune Response," *PLOS One*, 2014, 9(6): e99887. cited by applicant

International Preliminary Report of Patentability mailed Oct. 31, 2019 and received in PCT/US2018/028753, 9 pages. cited by applicant

Nezafat et al., "Designing an efficient multi-epitope peptide vaccine against *Vibrio cholerae* via combined immunoinformatics and protein interaction based approaches," *Comput Biol Chem*, 2016, 62:82-95. cited by applicant

Official foreign office action issued for the corresponding EP Patent Application No. 18726576.4 on Nov. 16, 2020, 6 pages. cited by applicant

Wilke et al., "Crystal Structure of the conserved domain of the DC lysosomal associated membrane protein: implications for the lysosomal glycocalyx," *BMC Biol.*, Oct. 2012:1-15. cited by applicant

Wimer-Mackin et al., "Transmembrane Domain Mutations Influence the Cellular Distribution of

Lysosomal Membrane Glycoprotein A,” Biochemical and Biophysical Research Comm., 1996, 229(2):472-478. cited by applicant
Written Opinion and International Search Report mailed Jun. 26, 2018 in PCT/US2018/028753, 16 pages. cited by applicant
Zhang et al., “Enhancement of Antitumor Immunity Using a DNA-Based Replicon Vaccine Derived from Semliki Forest Virus,” PLOS One, 2014, 9(3): e90551. cited by applicant
Zhou et al., “Lamp-2a Facilitates MHC Class II Presentation of Cytoplasmic Antigens,” Immunity, 2005, 22(5):571-581. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a US National Stage application of PCT/US19/32305 filed May 14, 2019, which claims priority is U.S. Provisional Application 62/672,005 filed on May 15, 2018, U.S. Provisional Application 62/672,378 filed on May 16, 2018, and U.S. Provisional Application 62/673,932 filed on May 20, 2018; all of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

Sequence Listing

(1) The present application contains a Sequence Listing which has been submitted in electronic format. The Sequence Listing, created on Dec. 20, 2023, is named “2023-12-20_01305-0013-OOUS_SL_Updated_ST25.txt” and is 639,432 bytes in size. The information in the electronic format of the Sequence Listing is incorporated herein by reference in its entirety.

Field of the Invention

(2) The invention relates to improved LAMP Constructs comprising allergens and their use in treating subjects suffering from allergic reactions and/or allergies. More specifically, the invention relates to nucleic acids for use as DNA vaccines, and methods of using them to treat subjects suffering from or susceptible to allergic reactions. Prime boost protocols utilizing the improved LAMP Constructs described herein are also described.

DISCUSSION OF THE RELATED ART

(3) In the following discussion, certain articles and methods will be described for background and introductory purposes. Nothing contained herein is to be construed as an “admission” of prior art. Applicant expressly reserves the right to demonstrate, where appropriate, that the articles and methods referenced herein do not constitute prior art under the applicable statutory provisions.

(4) Allergic reactions occur when the immune system reacts to harmless foreign substances, called allergens. For example, food allergies are an important public health issue due to the high risk of anaphylaxis, a potentially deadly systemic shock (Sampson et al. (1992) N. Engl. J. Med. 327:380-384; Bock et al. (2001) J. Allergy Clin. Immunol. 107:191-193). Young children are at greater risk of developing food allergies than the general public (Lack et al. (2003) N. Engl. J. Med. 348:977-985; Zimmerman et al. (1989) J. Allergy Clin. Immunol. 83:764-770; Green et al. (2007) Pediatrics 120:1304-1310). During the first three years of life, 6-8% of children experience an allergic reaction caused by food (Bock (1987) Allergy 45:587-596; Burks and Sampson (1993) Curr. Prob. Pediatr. 23:230-252; Jansen et al. (1994) J. Allergy Clin. Immunol. 93; 2:446-456; Sampson (1999) J. Allergy Clin. Immunol. 103; 5:717-728). Nut allergies, e.g., peanut and nut allergies, affect up to 1-2% of the population, and the rate of occurrence of this food allergy is thought to be increasing in

the general population, disproportionately affecting those of Asian ethnicity.

(5) Anaphylaxis caused by exposure to an allergen, e.g., tree nuts or peanuts, results in a severe immune reaction characterized by overproduction of histamine and is responsible for half of U.S. anaphylaxis emergency room visits annually. For example, extreme reactions to nuts result in over 30,000 incidents of anaphylaxis and between 100-200 deaths in the U.S. each year. Nuts in trace amounts are commonly found in thousands of individually branded, but not labeled, packaged food items. More than one and a half million Americans suffer symptoms from nut allergy and symptoms often persist throughout life. Many experience dangerous reactions on exposure to trace amounts.

(6) There is no treatment for relieving nut allergy symptoms. Over the last ten years, the prevalence of nut allergies has doubled to affect 2% of adult Americans (Sampson (1999) *J. Allergy Clin. Immunol.* 103; 5:717-728; Sicherer et al. (2003) *J. Allergy Clin. Immunol.* 112:1203-1207). While the symptoms for many other allergies like hay fever and short ragweed pollen are not life threatening, for a nut allergic individual, the ingestion of as little as 1/1000th of a nut can induce anaphylactic shock and death (Taylor et al. (2002) *J. Allergy Clin. Immunol.* 109 (1):24-30; Wensing et al. (2002) *J. Allergy Clin. Immunol.* 110(6):915-920). In the event that accidental ingestion triggers anaphylaxis, injections of epinephrine are used to open up airway passages (Stark and Sullivan (1986) *J. Allergy Clin. Immunol.* 78:76-83; Sampson (2003) *Pediatrics* 111(6):1601-1608).

(7) Food allergies occur when an individual fails to develop oral tolerance and instead becomes sensitized to subsequent allergen exposure (Till et al. (2004) *J. Allergy Clin. Immunol.* 113(6):1025-1034). In allergic patients, allergens preferentially activate type 2 helper CD4+T lymphocytes (Th2), which produce the pro-allergic cytokines interleukin IL-4, IL-5, and IL-13 that help orchestrate inflammation underlying most allergic symptoms (Woodfolk (2007) *J. Allergy Clin. Immunol.* 118(2):260-294). IL-4 instructs antibody-producing B cells to secrete allergen-specific Immunoglobulin (Ig) E (Del Prete et al. (1988) *J. Immunol.* 140:4193-4198; Swain et al. (1990) *J. Immunol.* 145:3796-3806). Unlike neutralizing IgG, IgE binds to its high affinity receptor Fc-εR1 expressed by mast cells and eosinophils (Blank et al. (1989) *Nature* 337:187-190; Benhamou et al. (1990) *J. Immunol.* 144:3071-3077), thus sensitizing these cells. Upon subsequent exposure, IgE binds the offending allergen, cross-links, and transduces a signal instructing mast cells to degranulate and release the volatile chemicals that trigger the allergic reaction.

(8) Beside food allergies, other environmental agents can also generate an allergic response as described above in an individual. Examples of such environmental agents include, but are not limited to, pollen, dog dander, cat saliva, or dust mites.

(9) Immunotherapy, the administration of increasing doses of an allergen to bring about tolerance, is a standard treatment for allergic diseases, but has not been approved for treating nut allergies due to frequent anaphylactic reactions (Nelson et al. (1997) *J. Allergy Clin. Immunol.* 99; 6:744-751; Oppenheimer et al. (1992) *J. Allergy Clin. Immunol.* 90:256-262). In addition, the utility of immunotherapy is limited by the length of treatment, which requires up to 36 months of weekly or bi-weekly injections and results in varying degrees of success and compliance (Bousquet et al. (1998) *J. Allergy Clin. Immunol.* 102:558-562; Rank and Li (2007) *Mayo Clin. Proc.* 82(9):1119-1123; Ciprandi et al. (2007) *Allergy Asthma Proc.* 28:40-43).

(10) DNA vaccines have been proposed as a treatment of allergic disease (Raz et al., 1996; Hartl et al., 2004; Hsu et al., 1996; Cramer 2007; Weiss et al., 2006). The underlying rationale is that allergen protein encoded by a DNA vaccine will preferentially activate the allergen-specific Th1 cellular response with the production of interferons by APCs, natural killer (NK), and T cells, rather than the characteristic Th2-type response, such as secretion of IL-4, IL-5, and IL-13, and the formation of IgE by B lymphocytes and the maturation and recruitment of eosinophils in late-phase reactions. However, the mechanisms underlying the differential induction of the Th1 and Th2 T-cell phenotypes appear to involve a large number of factors, such as unique properties of the bacterial

DNA of vaccine preparations, e.g., unmethylated and CpG DNA residues, the cytokine milieu elicited by innate immunity, and the cellular trafficking properties of the allergens (Chen et al., 2001; Kaech et al., 2002).

(11) DNA vaccines are new and promising candidates for the development of both prophylactic and therapeutic vaccines. They are proven to be safe and the lack of immune responses to a vector backbone may be a definitive advantage if repetitive cycles of vaccination are required to achieve clinical benefits. However, one perceived disadvantage of conventional DNA vaccines is their low immunogenicity in humans. A key limiting step in the immunogenicity of epitope-based DNA vaccines may be the access of epitopes to the MHCII presentation pathway to T cells, which is likely a stochastic process in the case of a vaccine without targeting technology.

(12) U.S. Pat. No. 5,633,234 describes chimeric proteins comprising an antigenic domain of modified influenza hemagglutinin (HA) and a cytoplasmic endosomal/lysosomal targeting signal which effectively target antigens to that compartment. The antigenic domain was processed and peptides from it presented on the cell surface in association with major histocompatibility (MHC) class II molecules. The cytoplasmic tail of LAMP-1 was used to form the endosomal/lysosomal targeting domain of the chimeric protein.

(13) U.S. Pat. No. 8,318,173 extended these initial observations to describe chimeric proteins (and the corresponding DNAs that encode these proteins) comprising the HIV-1 Gag protein inserted between the full luminal domain and a transmembrane domain of LAMP-1. This construct was introduced into dendritic cells which were then reported to target the MHC II pathway.

(14) This approach has proved useful in increasing cellular and humoral responses to several virus antigens, human papillomavirus E7, dengue virus membrane protein, HIV-1 gp160 membrane protein, HIV-1 p55 Gag, West Nile membrane protein, hepatitis C virus NS3 protein and cytomegalovirus pp65 (see, e.g., Bonini, et al., J. Immunol. 166: 5250-5257, 2001). The enhanced immune response can be attributed to co-localization of LAMP with MHC II and the more efficient processing and delivery of antigenic peptides. In addition, LAMP-targeting is reported to result in the presentation of an increased number of immunogenic epitopes, thus inducing a qualitatively broadened immune response compared to untargeted antigen. For example, Fernandes et al., 2000, Eur. J. Immunol. 30(8): 2333-43, demonstrated an increase in the number of presented peptides of a LAMP-trafficked OVA antigen encoded in a vaccinia vector. Of 12 peptides generated from exogenously supplied OVA, 9 were presented by an OVA/LAMP chimera, as compared to only 2 by the construct without LAMP.

(15) While it has been determined that the cytoplasmic domain of LAMP is necessary (in conjunction with a signal sequence and transmembrane domain), it is not always sufficient for endosomal/lysosomal trafficking of all antigens. Instead, the full luminal domain of LAMP has been shown to be also required for the trafficking of proteins to the lysosomal vesicular pathway.

(16) However, even with the presence of the complete luminal domain and the complete transmembrane/cytoplasmic tail of LAMP ("complete LAMP Constructs"), it has increasingly been found that the efficacy of a particular antigen to raise an immune response is highly dependent on the particular sequence used in these constructs. In fact, different antigenic fragments of the same protein when inserted into the complete LAMP constructs have been found to not elicit the same immune response. Sometimes the antigen fragment generates an immune response and other times it does not. These observations make the ability to predict ahead of time which particular antigenic sequence from a protein of interest will raise an immune response difficult with the complete LAMP Constructs.

(17) Moreover, in generating the complete LAMP Constructs, it has been repeatedly observed that the full luminal domain is required to properly express and process an antigen. For example, in Godinho et al., PLoS ONE 9(6): 9(6): e99887. doi:10.1371/journal.pone.0099887, the authors reported that the complete and intact luminal domain was the necessary minimal region needed to target an antigen to the lysosomes and that fragments of the luminal domain did not work. See, id.

at page 6.

(18) Specifically, the Godinho authors showed that by completely removing the first luminal domain and some of the second luminal domain (i.e., T1-Lum/gag construct), both protein expression and antibody response is decreased. Similarly, removing 25% of first luminal domain but having an intact second luminal domain (i.e., T2-lum/gag), both protein expression and antibody response comparatively increased but still less than the results obtained with the complete LAMP construct.

(19) Moreover, the authors acknowledged that the ability to raise an immune response is dependent upon the particular antigen and the epitopes used in these complete LAMP Constructs. For example, on page 9, column 2, the authors state “accordingly, previous studies demonstrated that DNA vaccines that generate Gag secreted as VLP, or in a soluble form, induce different levels of T and B cell activation, which were also different from the response induced by cytoplasmic Gag.” Moreover, insertion of an antigenic sequence between the full luminal domain of LAMP and the full transmembrane/cytoplasmic domain of LAMP as has been described in the literature can result in such large polynucleotide sequences that it can become either too costly to produce at commercial levels or impractical from a scientific perspective.

(20) Thus, there is a need to design new and improved LAMP Constructs that can be used as vaccines to effectively treat, for example, allergic reactions and/or allergies. Moreover, once improved, these new LAMP Constructs can be used to generate antibodies directed to the allergens as described herein.

SUMMARY OF THE INVENTION

(21) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other features, details, utilities, and advantages of the claimed subject matter will be apparent from the following written Detailed Description including those aspects illustrated in the accompanying drawings and defined in the appended claims.

(22) It is an object of this invention to provide novel constructs (“improved LAMP Constructs”) comprising specific fragments and/or variants of LAMP domains that effectively present the allergens specified herein to the immune system to generate an enhanced immune response. These improved LAMP Constructs effectively direct the allergens to the lysosomal/endosomal compartment where they are processed and presented to major histocompatibility complex (MHC) class II molecules so that helper T cells are preferentially stimulated and/or antibodies are generated.

(23) The improved LAMP Constructs and methods described herein may elicit an immune response in a subject. The immune response may be an immune response to the epitopes of the Allergen X (SEQ ID NO:Y) in the improved LAMP Construct (e.g., vaccine). Vaccines arm the immune system of the subject such that the immune system may detect and destroy that which contains the Allergen X (SEQ ID NO:Y) of the vaccines in the subject. The improved LAMP Constructs and methods described herein may elicit a Th1 immune response in the subject. Th1 immune responses may include secretion of inflammatory cytokines (e.g., IFN γ , TNF α) by a subset of immune cells (e.g., allergen specific T-cells).

(24) In some cases, the Allergen X (SEQ ID NO:Y) used in the improved LAMP Constructs and methods described herein may be recognized by the immune system of a subject to elicit a Th1 immune response and release Type I cytokines. The Th1 response may be initiated by the interaction between the epitope and the T-cell, more specifically, the major histocompatibility complex (MHC) expressed by the T-cell. For example, high affinity binding of an epitope to an MHC receptor may stimulate a Th1 response. MHC receptors may be at least one of a plurality of types of MHC receptors. The MHC receptors engaged on a T-cell may vary across individuals in a population.

(25) In some cases, the immune response is a Type 1 immune response. In some cases the immune response is characterized by a ratio of Type I cytokine production to Type II cytokine production that is greater than 1. In some cases, the immune response is characterized by a ratio of Type I cytokine production to Type II cytokine production that is less than 1. In some cases, the immune response is characterized by a ratio of IFN γ production to IL-10 production that is greater than 1. In some cases, the immune response is characterized by a ratio of IFN γ production to IL-10 production that is less than 1.

(26) Prime boost protocols are also contemplated. For example, the invention further provides a method for generating an immune response in a subject to an Allergen X (SEQ ID NO:Y), comprising priming the subject with an improved LAMP Construct comprising an Allergen X (SEQ ID NO:Y) as described herein followed by at least one boosting of the subject with the allergen or a related allergen (e.g., a second allergen derived from the same or highly similar protein sequence). Mixtures of allergens can be used in either or both the priming and the boosting step. Use of an improved LAMP Construct for the prime step followed by an Allergen X (SEQ ID NO:Y) boost step has been shown to significantly produce higher titers, indicating the power of LAMP in enhancing antibody response.

(27) The invention further provides a nucleic acid molecule encoding any of the improved LAMP Constructs comprising the Allergen X (SEQ ID NO:Y) described herein. The improved LAMP Construct can comprise a nucleic acid wherein the nucleic acid molecule is operably linked to an expression control sequence. In one preferred aspect, the improved LAMP Construct is a vaccine vector, suitable for vaccinating a patient. In another aspect, the invention provides a delivery vehicle comprising the improved LAMP Construct for facilitating the introduction of the nucleic acid molecule encoding the allergen into a cell. The delivery vehicle may be lipid-based (e.g., a liposome formulation), viral-based (e.g., comprising viral proteins encapsulating the nucleic acid molecule), or cell-based.

(28) In preferred embodiments, the invention provides an injectable composition comprising an improved LAMP Construct comprising an Allergen X (SEQ ID NO:Y) of interest for eliciting an immune response (e.g., generation of antibodies) in a mammal to the allergen. In preferred embodiments, this vaccine generates a preferential Th1 response to a Th2 response. The improved LAMP Constructs comprise at least one epitope of an Allergen X (SEQ ID NO:Y) as described herein.

(29) The invention also provides a cell comprising any of the improved LAMP Constructs described herein. In one aspect, the cell is an antigen presenting cell. The antigen presenting cell may be a professional antigen presenting cell (e.g., a dendritic cell, macrophage, B cell, and the like) or an engineered antigen presenting cell (e.g., a non-professional antigen presenting cell engineered to express molecules required for antigen presentation, such as MHC class II molecules). The molecules required for antigen presentation may be derived from other cells, e.g., naturally occurring, or may themselves be engineered (e.g. mutated or modified to express desired properties, such as higher or lower affinity for an allergenic epitope). In one aspect, the antigen presenting cell does not express any co-stimulatory signals.

(30) The invention additionally provides a kit comprising a plurality of cells comprising any of the improved LAMP Constructs described herein. At least two of the cells express different MHC class II molecules, and each cell comprises the same LAMP Construct. In one aspect, a kit is provided comprising an improved LAMP Construct and a cell for receiving the vector.

(31) The invention also provides a transgenic animal comprising at least one of the cells and/or at least one of the improved LAMP Constructs described herein. The invention also provides a transgenic animal comprising at least one of the cells described herein.

(32) The invention further provides a method for generating an immune response in an animal (e.g., a human or a non-human vertebrate) to an Allergen X (SEQ ID NO:Y), comprising: administering to the animal a cell as described above, wherein the cell expresses, or can be induced to express,

the improved LAMP Construct in the animal. In one aspect, the cell comprises an MHC class II molecule compatible with MHC proteins of the animal, such that the animal does not generate an immune response against the MHC class II molecule. In one preferred aspect, the animal is a human.

(33) In one further aspect, the invention provides a method for eliciting an immune response to an Allergen X (SEQ ID NO:Y), comprising administering to an animal, such as a human or a non-human vertebrate, any of the improved LAMP Constructs described herein. Preferably, the improved LAMP Construct is infectious for a cell of the animal. For example, the improved LAMP Construct may be a viral vector, such as a vaccinia improved LAMP Construct.

(34) For example, the invention further provides a method for generating an immune response in an animal to an Allergen X (SEQ ID NO:Y), comprising priming the animal with an improved LAMP Construct comprising an Allergen X (SEQ ID NO:Y) as described herein followed by at least one boosting of the animal. Use of an improved LAMP Construct for the prime step followed by an Allergen X (SEQ ID NO:Y) boost step has been shown to significantly produce higher titers, indicating the power of LAMP in enhancing antibody response.

(35) In a further aspect, a cell is obtained from a patient, the improved LAMP Construct described herein is introduced into the cell and the cell or progeny of the cell is reintroduced into the patient. In one aspect, the cell is a stem cell-capable of differentiating into an antigen presenting cell. Treatments of human patients as well as veterinary use are specifically contemplated.

(36) Specifically, by combining presentation of the allergen of interest with LAMP, the allergen is then effectively transported to the cytoplasmic endosomal/lysosomal compartments, where the allergen can be processed and peptides from it presented on the cell surface in association with major histocompatibility (MHC) class II molecules.

(37) These and other aspects, objects and features are described in more detail below.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The objects and features of the invention can be better understood with reference to the following detailed description and accompanying drawings.

(2) FIG. 1 illustrates the general scheme of different types of improved LAMP Constructs (identified as ILC-1, ILC-2, ILC-3, ILC-4, ILC-5 and ILC-6) that can be used as described herein.

(3) FIG. 2B illustrates the domains of the LAMP proteins defined herein while FIG. 2A defines the specific amino acid boundaries of these domains for human LAMP-1 (SEQ ID NO: 1), human LAMP-2 (SEQ ID NO:2), human LAMP-3 (SEQ ID NO:3), human LAMP-4 (SEQ ID NO:4), human Endolyn (SEQ ID NO:5), human Macrosailin (SEQ ID NO:80), human LAMP-5 (SEQ ID NO:93) and human LIMBIC (SEQ ID NO: 67). As described herein the LAMP luminal domains, homology domains, transmembrane domains, the cytoplasmic tail and the signal sequences can be used to generate the improved LAMP Constructs ILC-1, ILC-2, ILC-3, ILC-4, ILC-5 and ILC-6 as described herein.

(4) FIG. 3 provides alignment of LAMP-1 proteins found in other species as compared to human LAMP-1 (SEQ ID NO: 1). The equivalent domains of these other species can be used to generate the improved LAMP Constructs described herein and are readily identifiable by comparing the domains identified for human LAMP-1 in FIG. 2 and FIG. 3 to the alignments shown in FIG. 3.

(5) FIG. 4 provides alignment of LAMP-2 proteins found in other species as compared to human LAMP-2 (SEQ ID NO:2). The equivalent domains of these other species can be used to generate the improved LAMP Constructs described herein and are readily identifiable by comparing the domains identified for human LAMP-2 in FIG. 2 and FIG. 4 to the alignments shown in FIG. 4.

(6) FIG. 5 provides alignment of LAMP-3 proteins found in other species as compared to human

LAMP-3 (SEQ ID NO:3). The equivalent domains of these other species can be used to generate the improved LAMP Constructs described herein and are readily identifiable by comparing the domains identified for human LAMP-3 in FIG. 2 and FIG. 5 to the alignments shown in FIG. 5. (7) FIG. 6 provides alignment of LIMP-2 proteins found in other species as compared to human LIMP-2 (SEQ ID NO:4). The equivalent domains of these other species can be used to generate the improved LAMP Constructs described herein and are readily identifiable by comparing the domains identified for human LIMP-2 in FIG. 2 and FIG. 6 to the alignments shown in FIG. 6. (8) FIG. 7 provides alignment of LIMBIC proteins found in other species as compared to human LIMBIC (SEQ ID NO:67). The equivalent domains of these other species can be used to generate the improved LAMP Constructs described herein and are readily identifiable by comparing the domains identified for human LIMBIC in FIG. 2 and FIG. 7 to the alignments shown in FIG. 7. (9) FIG. 8 provides alignment of Endolyn proteins found in other species as compared to human Endolyn (SEQ ID NO:5). The equivalent domains of these other species can be used to generate the improved LAMP Constructs described herein and are readily identifiable by comparing the domains identified for human Endolyn in FIG. 2 and FIG. 8 to the alignments shown in FIG. 8. (10) FIG. 9 provides alignment of Macrosailin proteins found in other species as compared to human Macrosailin (SEQ ID NO:80). The equivalent domains of these other species can be used to generate the improved LAMP Constructs described herein and are readily identifiable by comparing the domains identified for human Macrosailin in FIG. 2 and FIG. 9 to the alignments shown in FIG. 9.

(11) FIG. 10 provides alignment of LAMP-5 proteins found in other species as compared to human LAMP-5 (SEQ ID NO:93). The equivalent domains of these other species can be used to generate the improved LAMP Constructs described herein and are readily identifiable by comparing the domains identified for human LAMP-5 in FIG. 2 and FIG. 10 to the alignments shown in FIG. 10.

(12) FIG. 11 shows results obtained when mice were immunized with HVEM-LAMP, HVEM, or LAMP on day 0, 7, and 14. On day 28, mice were bled and serum samples were isolated. HVEM specific IgG was examined by ELISA. Data represent geometric mean of antibody titers $\pm$ geometric SD, n=6. ** p value <0.01

(13) FIG. 12 shows results obtained when mice were immunized with HVEM-LAMP, HVEM, or LAMP on day 0, 7, and 14. On day 35, mice were boosted with 5 μ g HVEM protein in the presence of alum adjuvant. Mice were bled on day 49 and serum samples were isolated. HVEM specific IgG was examined by ELISA. Data represent geometric mean of antibody titers $\pm$ geometric SD, n=6. *** p value <0.001; **** p value <0.0001.

(14) FIG. 13 shows that LAMP alters the binding affinity of epitopes in CRD3/4 of HVEM.

(15) FIG. 14 is a schematic representation of the different constructs described herein as provided as Table 1/FIG. 14.

(16) FIG. 15A and FIG. 15B is a Western Blot of allergens Amb a 1, Bet v 1, Fel d 4 expressed from different LAMP constructs and FIG. 15C is a Western Blot of allergen Cry J1/Cry J2. Protein expression for each allergen was demonstrated with anti-LAMP antibody (FIG. 15A and FIG. 15C). GAPDH detection was used to demonstrate equal loading between lanes (FIG. 15B).

(17) FIG. 16 shows the results of Amb a 1 specific IgG ELISA.

(18) FIG. 17 shows the results of Bet v 1 specific IgG ELISA.

(19) FIG. 18 shows the results of Fel d 4 specific IgG ELISA

(20) FIG. 19 shows the results of Cry J 1 specific IgG ELISA.

DETAILED DESCRIPTION

(21) The invention provides improved LAMP Constructs which can be used to generate vaccines. The improved LAMP Constructs can be used to modulate or enhance an immune response. In one preferred aspect, the invention provides a method for treating a patient with allergies and/or an allergic response by providing an improved LAMP Construct comprising one or more of the Allergen X (SEQ ID NO.Y) as described herein.

(22) Definitions

(23) The following definitions are provided for specific terms which are used in the following written description.

(24) As used in the specification and claims, the singular form “a”, “an” and “the” include plural references unless the context clearly dictates otherwise. For example, the term “a cell” includes a plurality of cells, including mixtures thereof. The term “a nucleic acid molecule” includes a plurality of nucleic acid molecules.

(25) As used herein, the term “comprising” is intended to mean that the improved LAMP Constructs and methods include the recited elements, but do not exclude other elements.

“Consisting essentially of”, when used to define improved LAMP Constructs and methods, shall mean excluding other elements of any essential significance to the combination. Thus, an improved LAMP Construct consisting essentially of the elements as defined herein would not exclude trace contaminants from the isolation and purification method and pharmaceutically acceptable carriers, such as phosphate buffered saline, preservatives, and the like. “Consisting of” shall mean excluding more than trace elements of other ingredients and substantial method steps for administering the improved LAMP Constructs of this invention. Embodiments defined by each of these transition terms are within the scope of this invention.

(26) The term “about” or “approximately” means within an acceptable range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, e.g., the limitations of the measurement system. For example, “about” can mean a range of up to 20%, preferably up to 10%, more preferably up to 5%, and more preferably still up to 1% of a given value. Alternatively, particularly with respect to biological systems or processes, the term can mean within an order of magnitude, preferably within 5 fold, and more preferably within 2 fold, of a value. Unless otherwise stated, the term ‘about’ means within an acceptable error range for the particular value, such as ± 1 -20%, preferably ± 1 -10% and more preferably ± 1 -5%.

(27) Where a range of values is provided, it is understood that each intervening value, between the upper and lower limit of that range and any other stated or intervening value in that stated range is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges, and are also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either both of those included limits are also included in the invention.

(28) As used herein, “the lysosomal/endosomal compartment” refers to membrane-bound acidic vacuoles containing LAMP molecules in the membrane, hydrolytic enzymes that function in antigen processing, and MHC class II molecules for antigen recognition and presentation. This compartment functions as a site for degradation of foreign materials internalized from the cell surface by any of a variety of mechanisms including endocytosis, phagocytosis and pinocytosis, and of intracellular material delivered to this compartment by specialized autolytic phenomena (de Duve, Eur. J. Biochem. 137: 391, 1983). The term “endosome” as used herein and in the claims encompasses a lysosome.

(29) As used herein, a “lysosome-related organelle” refers to any organelle which comprises lysosomes and includes, but is not limited to, MIIC, CIIV, melanosomes, secretory granules, lytic granules, platelet-dense granules, basophil granules, Birbeck granules, phagolysosomes, secretory lysosomes, and the like. Preferably, such an organelle lacks mannose 6-phosphate receptors and comprises LAMP, but may or may not comprise an MHC class II molecule. For reviews, see, e.g., Blott and Griffiths, Nature Reviews, Molecular Cell Biology, 2002; Dell’Angelica, et al., The FASEB Journal 14: 1265-1278, 2000.

(30) As used herein, the terms “polynucleotide” and “nucleic acid molecule” are used interchangeably to refer to polymeric forms of nucleotides of any length. The polynucleotides may

contain deoxyribonucleotides, ribonucleotides, and/or their analogs. Nucleotides may have any three-dimensional structure, and may perform any function, known or unknown. The term “polynucleotide” includes, for example, single-, double-stranded and triple helical molecules, a gene or gene fragment, exons, introns, mRNA, tRNA, rRNA, ribozymes, antisense molecules, cDNA, recombinant polynucleotides, branched polynucleotides, aptamers, plasmids, vectors, isolated DNA of any sequence, isolated RNA of any sequence, nucleic acid probes, and primers. A nucleic acid molecule may also comprise modified nucleic acid molecules (e.g., comprising modified bases, sugars, and/or internucleotide linkers).

(31) As used herein, the term “peptide” refers to a compound of two or more subunit amino acids, amino acid analogs, or peptidomimetics. The subunits may be linked by peptide bonds or by other bonds (e.g., as esters, ethers, and the like).

(32) As used herein, the term “amino acid” refers to either natural and/or unnatural or synthetic amino acids, including glycine and both D or L optical isomers, and amino acid analogs and peptidomimetics. A peptide of three or more amino acids is commonly called an oligopeptide if the peptide chain is short. If the peptide chain is long (e.g., greater than about 10 amino acids), the peptide is commonly called a polypeptide or a protein. While the term “protein” encompasses the term “polypeptide”, a “polypeptide” may be a less than full-length protein.

(33) As used herein a “LAMP polypeptide” refers to the mammalian lysosomal associated membrane proteins human LAMP-1, human LAMP-2, human LAMP-3, human LIMP-2, human Endolyn, human LIMBIC, human LAMP-5, or human Macrosailin as described herein, as well as orthologs (such as, for example, the LAMP proteins shown in FIGS. 3-10), and allelic variants. As used herein, “expression” refers to the process by which polynucleotides are transcribed into mRNA and/or translated into peptides, polypeptides, or proteins. If the polynucleotide is derived from genomic DNA, expression may include splicing of the mRNA transcribed from the genomic DNA.

(34) As used herein, “under transcriptional control” or “operably linked” refers to expression (e.g., transcription or translation) of a polynucleotide sequence which is controlled by an appropriate juxtaposition of an expression control element and a coding sequence. In one aspect, a DNA sequence is “operatively linked” to an expression control sequence when the expression control sequence controls and regulates the transcription of that DNA sequence.

(35) As used herein, “coding sequence” is a sequence which is transcribed and translated into a polypeptide when placed under the control of appropriate expression control sequences. The boundaries of a coding sequence are determined by a start codon at the 5' (amino) terminus and a translation stop codon at the 3' (carboxyl) terminus. A coding sequence can include, but is not limited to, a prokaryotic sequence, cDNA from eukaryotic mRNA, a genomic DNA sequence from eukaryotic (e.g., mammalian) DNA, and even synthetic DNA sequences. A polyadenylation signal and transcription termination sequence will usually be located 3' to the coding sequence.

(36) As used herein, two coding sequences “correspond” to each other if the sequences or their complementary sequences encode the same amino acid sequences.

(37) As used herein, “signal sequence” denotes the endoplasmic reticulum translocation sequence. This sequence encodes a signal peptide that communicates to a cell to direct a polypeptide to which it is linked (e.g., via a chemical bond) to an endoplasmic reticulum vesicular compartment, to enter an exocytic/endocytic organelle, to be delivered either to a cellular vesicular compartment, the cell surface or to secrete the polypeptide. This signal sequence is sometimes clipped off by the cell in the maturation of a protein. Signal sequences can be found associated with a variety of proteins native to prokaryotes and eukaryotes.

(38) As used herein, “trafficking” denotes movement or progression of the polypeptide encoded by the improved LAMP Construct through cellular organelles or compartments in the pathway from the rough endoplasmic reticulum to the endosomal/lysosomal compartment or related organelles where antigen processing and binding to MHC II occurs.

(39) Short stretches of polynucleotides that encode amino acids may be included at the ends of the polynucleotides that encode Allergen X to facilitate cloning into any of the vectors described herein. For example, use of cloning sequences that flank the amino acid sequence of SEQ ID NO:Y, such as, for example, polynucleotides that encode “Leu-Glu” and “Glu-Phe” (e.g., “CTCGAG” and “GAATTC”) may be included in the construct design.

(40) As used herein, an “improved LAMP Construct” and an “improved LAMP Construct comprising an Allergen X (SEQ ID NO:Y)” and an “improved LAMP Construct comprising an allergen of interest” are used interchangeably. The different arrangements of the improved LAMP Constructs are illustrated in FIG. 1 as ILC1-ILC6. Moreover, the use of an “improved LAMP Construct” encompasses both the polynucleotide sequence of the improved LAMP Construct (which comprises a polynucleotide encoding Allergen X (SEQ ID NO:Y) as well as the protein encoded by the polynucleotide sequence of the improved LAMP Construct.

(41) As used herein, an “improved LAMP Construct delivery vehicle” is defined as any molecule or group of molecules or macromolecules that can carry an improved LAMP Construct into a host cell (e.g., such as genes or gene fragments, antisense molecules, ribozymes, aptamers, and the like) and which occurs in association with an improved LAMP Construct as described herein.

(42) As used herein, “improved LAMP Construct delivery,” or “improved LAMP Construct transfer,” refers to the introduction of the improved LAMP Construct into a host cell, irrespective of the method used for the introduction. The introduced improved LAMP Constructs may be stably or transiently maintained in the host cell. Stable maintenance typically requires that the introduced improved LAMP Construct either contains an origin of replication compatible with the host cell or integrates into a replicon of the host cell such as an extrachromosomal replicon (e.g., a plasmid) or a nuclear or mitochondrial chromosome.

(43) As used herein, a “viral improved LAMP Construct” refers to a virus or viral particle that comprises the improved LAMP Construct to be delivered into a host cell, either in vivo, ex vivo or in vitro. Examples of viral improved LAMP Constructs include, but are not limited to, adenovirus vectors, adeno-associated virus vectors, retroviral vectors, and the like. In aspects where gene transfer is mediated by an adenoviral vector, an improved LAMP Construct includes the adenovirus genome or part thereof, and a selected, non-adenoviral gene, in association with adenoviral capsid proteins.

(44) As used herein, “adenoviral-mediated gene transfer” or “adenoviral transduction” refers to the process by which an improved LAMP Construct is transferred into a host cell by virtue of the adenovirus entering the cell. Preferably, the improved LAMP Construct is able to replicate and/or integrate and be transcribed within the cell.

(45) As used herein, “adenovirus particles” are individual adenovirus virions comprised of an external capsid and an improved LAMP Construct, where the capsid is further comprised of adenovirus envelope proteins. The adenovirus envelope proteins may be modified to comprise a fusion polypeptide which contains a polypeptide ligand covalently attached to the viral protein, e.g., for targeting the adenoviral particle to a particular cell and/or tissue type.

(46) As used herein, the term “administering” or “immunizing” or “injecting” an improved LAMP Construct refers to transducing, transfecting, microinjecting, electroporating, or shooting the cell with the improved LAMP Construct. In some aspects, improved LAMP Constructs are introduced into a target cell by contacting the target cell with a delivery cell (e.g., by cell fusion or by lysing the delivery cell when it is in proximity to the target cell).

(47) As used herein, the phrase “prime boost” describes the use of an improved LAMP Construct comprising an Allergen X (SEQ ID NO:Y) described herein used to prime a T-cell response followed by the use of a second improved LAMP Construct comprising an Allergen X (SEQ ID NO:Y), a DNA vaccine comprising an Allergen X (SEQ ID NO:Y) or a recombinant allergen to boost the response (or visa versa). These heterologous prime-boost immunizations elicit immune responses of greater height and breadth than can be achieved by priming and boosting with the

same vector. The priming with an improved LAMP Construct comprising an Allergen X (SEQ ID NO:Y) initiates memory cells; the boost step expands the memory response. Preferably, two different agents that do not raise responses against each other are used and thus do not interfere with each other's activity. Mixtures of allergens are specifically contemplated in the prime and/or boost step. Boosting can occur once or multiple times.

(48) As used herein, “hybridization” refers to a reaction in which one or more polynucleotides react to form a complex that is stabilized via hydrogen bonding between the bases of the nucleotide residues. The hydrogen bonding may occur by Watson-Crick base pairing, Hoogsteen binding, or in any other sequence-specific manner. The complex may comprise two strands forming a duplex structure, three or more strands forming a multi-stranded complex, a single self-hybridizing strand, or any combination of these. A hybridization reaction may constitute a step in a more extensive process, such as the initiation of a PCR reaction, or the enzymatic cleavage of a polynucleotide by a ribozyme.

(49) As used herein, a polynucleotide or polynucleotide region (or a polypeptide or polypeptide region) which has a certain percentage (for example, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 99%) of “sequence identity” to another sequence means that, when maximally aligned, using software programs routine in the art, that percentage of bases (or amino acids) are the same in comparing the two sequences.

(50) Two sequences are “substantially homologous” or “substantially similar” when at least about 50%, at least about 60%, at least about 70%, at least about 75%, and preferably at least about 80%, and most preferably at least about 90 or 95% of the nucleotides match over the defined length of the DNA sequences. Similarly, two polypeptide sequences are “substantially homologous” or “substantially similar” when at least about 50%, at least about 60%, at least about 66%, at least about 70%, at least about 75%, and preferably at least about 80%, and most preferably at least about 90 or 95% of the amino acid residues of the polypeptide match over a defined length of the polypeptide sequence. Sequences that are substantially homologous can be identified by comparing the sequences using standard software available in sequence data banks. Substantially homologous nucleic acid sequences also can be identified in a Southern hybridization experiment under, for example, stringent conditions as defined for that particular system. Defining appropriate hybridization conditions is within the skill of the art. For example, stringent conditions can be: hybridization at 5×SSC and 50% formamide at 42° C., and washing at 0.1×SSC and 0.1% sodium dodecyl sulfate at 60° C. Further examples of stringent hybridization conditions include: incubation temperatures of about 25 degrees C. to about 37 degrees C.; hybridization buffer concentrations of about 6×SSC to about 10×SSC; formamide concentrations of about 0% to about 25%; and wash solutions of about 6×SSC. Examples of moderate hybridization conditions include: incubation temperatures of about 40 degrees C. to about 50 degrees C.; buffer concentrations of about 9×SSC to about 2×SSC; formamide concentrations of about 30% to about 50%; and wash solutions of about 5×SSC to about 2×SSC. Examples of high stringency conditions include: incubation temperatures of about 55 degrees C. to about 68 degrees C.; buffer concentrations of about 1×SSC to about 0.1×SSC; formamide concentrations of about 55% to about 75%; and wash solutions of about 1×SSC, 0.1×SSC, or deionized water. In general, hybridization incubation times are from 5 minutes to 24 hours, with 1, 2, or more washing steps, and wash incubation times are about 1, 2, or 15 minutes. SSC is 0.15 M NaCl and 15 mM citrate buffer. It is understood that equivalents of SSC using other buffer systems can be employed. Similarity can be verified by sequencing, but preferably, is also or alternatively, verified by function (e.g., ability to traffic to an endosomal compartment, and the like), using assays suitable for the particular domain in question.

(51) The terms “percent (%) sequence similarity”, “percent (%) sequence identity”, and the like, generally refer to the degree of identity or correspondence between different nucleotide sequences of nucleic acid molecules or amino acid sequences of polypeptides that may or may not share a

common evolutionary origin (see Reeck et al., supra). Sequence identity can be determined using any of a number of publicly available sequence comparison algorithms, such as BLAST, FASTA, DNA Strider, GCG (Genetics Computer Group, Program Manual for the GCG Package, Version 7, Madison, Wisconsin), etc.

(52) To determine the percent identity between two amino acid sequences or two nucleic acid molecules, the sequences are aligned for optimal comparison purposes. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences (i.e., $\text{percent identity} = \frac{\text{number of identical positions}}{\text{total number of positions (e.g., overlapping positions)}} \times 100$). In one embodiment, the two sequences are, or are about, of the same length. The percent identity between two sequences can be determined using techniques similar to those described below, with or without allowing gaps. In calculating percent sequence identity, typically exact matches are counted.

(53) The determination of percent identity between two sequences can be accomplished using a mathematical algorithm. A non-limiting example of a mathematical algorithm utilized for the comparison of two sequences is the algorithm of Karlin and Altschul, Proc. Natl. Acad. Sci. USA 1990, 87:2264, modified as in Karlin and Altschul, Proc. Natl. Acad. Sci. USA 1993, 90:5873-5877. Such an algorithm is incorporated into the NBLAST and XBLAST programs of Altschul et al, J. Mol. Biol. 1990; 215: 403. BLAST nucleotide searches can be performed with the NBLAST program, score=100, wordlength=12, to obtain nucleotide sequences homologous to sequences of the invention. BLAST protein searches can be performed with the XBLAST program, score=50, wordlength=3, to obtain amino acid sequences homologous to protein sequences of the invention. To obtain gapped alignments for comparison purposes, Gapped BLAST can be utilized as described in Altschul et al, Nucleic Acids Res. 1997, 25:3389. Alternatively, PSI-Blast can be used to perform an iterated search that detects distant relationship between molecules. See Altschul et al. (1997) supra. When utilizing BLAST, Gapped BLAST, and PSI-Blast programs, the default parameters of the respective programs (e.g., XBLAST and NBLAST) can be used. See ncbi.nlm.nih.gov/BLAST/ on the WorldWideWeb.

(54) Another non-limiting example of a mathematical algorithm utilized for the comparison of sequences is the algorithm of Myers and Miller, CABIOS 1988; 4: 1 1-17. Such an algorithm is incorporated into the ALIGN program (version 2.0), which is part of the GCG sequence alignment software package. When utilizing the ALIGN program for comparing amino acid sequences, a PAM120 weight residue table, a gap length penalty of 12, and a gap penalty of 4 can be used.

(55) In a preferred embodiment, the percent identity between two amino acid sequences is determined using the algorithm of Needleman and Wunsch (J. Mol. Biol. 1970, 48:444-453), which has been incorporated into the GAP program in the GCG software package (Accelrys, Burlington, MA; available at accelrys.com on the WorldWideWeb), using either a Blossum 62 matrix or a PAM250 matrix, a gap weight of 16, 14, 12, 10, 8, 6, or 4, and a length weight of 1, 2, 3, 4, 5, or 6. In yet another preferred embodiment, the percent identity between two nucleotide sequences is determined using the GAP program in the GCG software package using a NWSgapdna.CMP matrix, a gap weight of 40, 50, 60, 70, or 80, and a length weight of 1, 2, 3, 4, 5, or 6. A particularly preferred set of parameters (and the one that can be used if the practitioner is uncertain about what parameters should be applied to determine if a molecule is a sequence identity or homology limitation of the invention) is using a Blossum 62 scoring matrix with a gap open penalty of 12, a gap extend penalty of 4, and a frameshift gap penalty of 5.

(56) Another non-limiting example of how percent identity can be determined is by using software programs such as those described in Current Protocols In Molecular Biology (F. M. Ausubel et al., eds., 1987) Supplement 30, section 7.7.18, Table 7.7.1. Preferably, default parameters are used for alignment. A preferred alignment program is BLAST, using default parameters. In particular, preferred programs are BLASTN and BLASTP, using the following default parameters: Genetic code=standard; filter=none; strand=both; cutoff=60; expect=10; Matrix=BLOSUM62;

Descriptions=50 sequences; sort by=HIGH SCORE; Databases=non-redundant, GenBank+EMBL+DDBJ+PDB+GenBank CDS translations+SwissProtein+SPupdate+PIR. Details of these programs can be found at the following Internet address: <http://www.ncbi.nlm.nih.gov/cgi-bin/BLAST>.

(57) Statistical analysis of the properties described herein may be carried out by standard tests, for example, t-tests, ANOVA, or Chi squared tests. Typically, statistical significance will be measured to a level of $p=0.05$ (5%), more preferably $p=0.01$, $p=0.001$, $p=0.0001$, $p=0.000001$

(58) “Conservatively modified variants” of domain sequences also can be provided. With respect to particular nucleic acid sequences, conservatively modified variants refer to those nucleic acids which encode identical or essentially identical amino acid sequences, or where the nucleic acid does not encode an amino acid sequence, to essentially identical sequences. Specifically, degenerate codon substitutions can be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed-base and/or deoxyinosine residues (Batzer, et al., 1991, *Nucleic Acid Res.* 19: 5081; Ohtsuka, et al., 1985, *J. Biol. Chem.* 260: 2605-2608; Rossolini et al., 1994, *Mol. Cell. Probes* 8: 91-98).

(59) The term “biologically active fragment”, “biologically active form”, “biologically active equivalent” of and “functional derivative” of a wild-type protein, possesses a biological activity that is at least substantially equal (e.g., not significantly different from) the biological activity of the wild type protein as measured using an assay suitable for detecting the activity.

(60) As used herein, “in vivo” nucleic acid delivery, nucleic acid transfer, nucleic acid therapy” and the like, refer to the introduction of an improved LAMP Construct directly into the body of an organism, such as a human or non-human mammal, whereby the improved LAMP Construct is introduced to a cell of such organism in vivo.

(61) As used herein, the term “in situ” refers to a type of in vivo nucleic acid delivery in which the improved LAMP Construct is brought into proximity with a target cell (e.g., the nucleic acid is not administered systemically). For example, in situ delivery methods include, but are not limited to, injecting an improved LAMP Construct directly at a site (e.g., into a tissue, such as a tumor or heart muscle), contacting the improved LAMP Construct with cell(s) or tissue through an open surgical field, or delivering the improved LAMP Constructs to a site using a medical access device such as a catheter.

(62) As used herein, the term “isolated” or “purified” means separated (or substantially free) from constituents, cellular and otherwise, in which the polynucleotide, peptide, polypeptide, protein, antibody, or fragments thereof, are normally associated with in nature. For example, with respect to an improved LAMP Construct, an isolated polynucleotide is one that is separated from the 5' and 3' sequences with which it is normally associated in the chromosome. As is apparent to those of skill in the art, a non-naturally occurring polynucleotide, peptide, polypeptide, protein, antibody, or fragments thereof, does not require “isolation” to distinguish it from its naturally occurring counterpart. By substantially free or substantially purified, it is meant at least 50% of the population, preferably at least 70%, more preferably at least 80%, and even more preferably at least 90%, are free of the components with which they are associated in nature.

(63) As used herein, a “target cell” or “recipient cell” refers to an individual cell or cell which is desired to be, or has been, a recipient of the improved LAMP Constructs described herein. The term is also intended to include progeny of a single cell, and the progeny may not necessarily be completely identical (in morphology or in genomic or total DNA complement) to the original parent cell due to natural, accidental, or deliberate mutation. A target cell may be in contact with other cells (e.g., as in a tissue) or may be found circulating within the body of an organism.

(64) As used herein, a “subject” is a vertebrate, preferably a mammal, more preferably a human. Mammals include, but are not limited to, murines, simians, humans, farm animals, sport animals, and pets. In other preferred embodiments, the “subject” is a rodent (e.g. a rat, a mouse, a rabbit, a llama, camels, a cow, a guinea pig, a hamster, a dog, a cat, a horse, a non-human primate, a simian

(e.g. a monkey or ape), a monkey (e.g. marmoset, baboon, rhesus macaque), or an ape (e.g. gorilla, chimpanzee, orangutan, gibbon). In other embodiments, non-human mammals, especially mammals that are conventionally used as models for demonstrating therapeutic efficacy in humans (e.g. murine, primate, porcine, canine, or rabbit animals) may be employed.

(65) As used herein, the term “pharmaceutically acceptable carrier” encompasses any of the standard pharmaceutical carriers, such as a phosphate buffered saline solution, water, and emulsions, such as an oil/water or water/oil emulsion, and various types of wetting agents. Compositions comprising the improved LAMP Constructs also can include stabilizers and preservatives. For examples of carriers, stabilizers and adjuvants, see Martin Remington's Pharm. Sci., 15th Ed. (Mack Publ. Co., Easton (1975)).

(66) A cell has been “transformed”, “transduced”, or “transfected” by the improved LAMP Constructs when such nucleic acids have been introduced inside the cell. Transforming DNA may or may not be integrated (covalently linked) with chromosomal DNA making up the genome of the cell. In prokaryotes, yeast, and mammalian cells for example, the improved LAMP Constructs may be maintained on an episomal element, such as a plasmid. In a eukaryotic cell, a stably transformed cell is one in which the improved LAMP Constructs have become integrated into a chromosome so that it is inherited by daughter cells through chromosome replication. This stability is demonstrated by the ability of the eukaryotic cell to establish cell lines or clones comprised of a population of daughter cells containing the improved LAMP Constructs. A “clone” is a population of cells derived from a single cell or common ancestor by mitosis. A “cell line” is a clone of a primary cell that is capable of stable growth in vitro for many generations (e.g., at least about 10).

(67) As used herein, an “effective amount” is an amount sufficient to affect beneficial or desired results, e.g., such as an effective amount of the improved LAMP Construct transfer and/or expression, and/or the attainment of a desired therapeutic endpoint. An effective amount can be administered in one or more administrations, applications or dosages. In one aspect, an effective amount of an improved LAMP Construct is an amount sufficient to transform/transduce/transfect at least one cell in a population of cells comprising at least two cells.

(68) As used herein, a “therapeutically effective amount” is used herein to mean an amount sufficient to prevent, correct and/or normalize an abnormal physiological response. In one aspect, a “therapeutically effective amount” is an amount sufficient to reduce by at least about 30 percent, more preferably by at least 50 percent, most preferably by at least 90 percent, a clinically significant feature of pathology, such as for example, size of a tumor mass, antibody production, cytokine production, fever or white cell count, etc.

(69) An “antibody” is any immunoglobulin, including antibodies and fragments thereof, that binds a specific antigen. The term encompasses polyclonal, monoclonal, and chimeric antibodies (e.g., bispecific antibodies). An “antibody combining site” is that structural portion of an antibody molecule comprised of heavy and light chain variable and hypervariable regions that specifically binds antigen. Exemplary antibody molecules are intact immunoglobulin molecules, substantially intact immunoglobulin molecules, and those portions of an immunoglobulin molecule that contains the paratope, including Fab, Fab', F(ab').sub.2 and F(v) portions, which portions are preferred for use in the therapeutic methods described herein. Thus, the term antibody encompasses not only whole antibody molecules, but also antibody fragments as well as variants (including derivatives such as fusion proteins) of antibodies and antibody fragments. Examples of molecules which are described by the term “antibody” in this application include, but are not limited to: single chain Fvs (scFvs), Fab fragments, Fab' fragments, F(ab').sub.2, disulfide linked Fvs (sdFvs), Fvs, and fragments comprising or alternatively consisting of, either a VL or a VH domain. The term “single chain Fv” or “scFv” as used herein refers to a polypeptide comprising a VL domain of an antibody linked to a VH domain of an antibody. See Carter (2006) Nature Rev. Immunol. 6:243.

(70) Additionally, antibodies of the invention include, but are not limited to, monoclonal, multi-specific, bi-specific, human, humanized, mouse, or chimeric antibodies, single chain antibodies,

camelid antibodies, Fab fragments, F(ab') fragments, anti-idiotypic (anti-Id) antibodies (including, e.g., anti-Id antibodies to antibodies of the invention), domain antibodies and epitope-binding fragments of any of the above. The immunoglobulin molecules of the invention can be of any type (e.g., IgG, IgE, IgM, IgD, IgA and IgY), class (e.g., IgG1, IgG2, IgG3, IgG4, IgA1 and IgA2) or subclass of immunoglobulin molecule.

(71) Most preferably, the antibodies are human antibodies. As used herein, "human" antibodies include antibodies having the amino acid sequence of a human immunoglobulin and include antibodies isolated from human immunoglobulin libraries and xenomice or other organisms that have been genetically engineered to produce human antibodies. The improved LAMP Constructs described herein can be used in combination with known techniques for generating human antibodies and human monoclonal antibodies as described in the exemplified protocols, see, e.g., PCT publications WO 98/24893; WO 92/01047; WO 96/34096; WO 96/33735; European Patent No. 0598877; U.S. Pat. Nos. 5,413,923; 5,625,126; 5,633,425; 5,569,825; 5,661,016; 5,545,806; 5,814,318; 5,885,793; 5,916,771; and 5,939,598; and Lonberg and Huszar, *Int. Rev. Immunol.* 13:65-93 (1995).

(72) Human antibodies or "humanized" chimeric monoclonal antibodies can be produced using the improved LAMP Constructs in combination with techniques described herein or otherwise known in the art. For example, standard methods for producing chimeric antibodies are known in the art. See, for review the following references: Morrison, *Science* 229:1202 (1985); Oi et al., *BioTechniques* 4:214 (1986); Cabilly et al., U.S. Pat. No. 4,816,567; Taniguchi et al., EP 171496; Morrison et al., EP 173494; Neuberger et al., WO 8601533; Robinson et al., WO 8702671; Boulianne et al., *Nature* 312:643 (1984); Neuberger et al., *Nature* 314:268 (1985).

(73) The antibodies of the present invention may be monovalent, bivalent, trivalent or multivalent. For example, monovalent scFvs can be multimerized either chemically or by association with another protein or substance. A scFv that is fused to a hexahistidine tag or a Flag tag can be multimerized using Ni-NTA agarose (Qiagen) or using anti-Flag antibodies (Stratagene, Inc.). Additionally, the improved LAMP Constructs can be used to generate monospecific, bispecific, trispecific or of greater multispecificity for the encoded allergen(s) contained in the improved LAMP Construct. See, e.g., PCT publications WO 93/17715; WO 92/08802; WO 91/00360; WO 92/05793; Tutt, et al., *J. Immunol.* 147:60-69 (1991); U.S. Pat. Nos. 4,474,893; 4,714,681; 4,925,648; 5,573,920; 5,601,819; Kostelny et. al., *J. Immunol.* 148:1547-1553 (1992).

(74) An "epitope" is a structure, usually made up of a short peptide sequence or oligosaccharide, that is specifically recognized or specifically bound by a component of the immune system. T-cell epitopes have generally been shown to be linear oligopeptides. Two epitopes correspond to each other if they can be specifically bound by the same antibody. Two epitopes correspond to each other if both are capable of binding to the same B cell receptor or to the same T cell receptor, and binding of one antibody to its epitope substantially prevents binding by the other epitope (e.g., less than about 30%, preferably, less than about 20%, and more preferably, less than about 10%, 5%, 1%, or about 0.1% of the other epitope binds). In the present invention, multiple epitopes can make up an Allergen X (SEQ ID NO:Y).

(75) The term "allergen" or "allergen of interest" as used herein covers any polypeptide sequence encoded by a polynucleotide sequence cloned into the improved LAMP Construct which is used to elicit an innate or adaptive immune response as shown in Table 1/FIG. 14. An "allergen" encompasses both a single allergen as well as multiple allergen sequences (derived from the same or different proteins) cloned into the improved LAMP Construct.

(76) The term "antigen presenting cell" as used herein includes any cell which presents on its surface an allergen in association with a major histocompatibility complex molecule, or portion thereof, or, alternatively, one or more non-classical MHC molecules, or a portion thereof. Examples of suitable APCs are discussed in detail below and include, but are not limited to, whole cells such as macrophages, dendritic cells, B cells, hybrid APCs, and foster antigen presenting cells.

(77) As used herein an “engineered antigen-presenting cell” refers to an antigen-presenting cell that has a non-natural molecular moiety on its surface. For example, such a cell may not naturally have a costimulator on its surface or may have an additional artificial costimulator in addition to a natural costimulator on its surface, or may express a non-natural class II molecule on its surface. In preferred embodiments, the engineered antigen-presenting cell has the allergen expressed from the improved LAMP Construct on its surface.

(78) As used herein, “immune effector cells” refers to cells capable of binding an allergen and which mediate an immune response. These cells include, but are not limited to, T cells, B cells, monocytes, macrophages, NK cells and cytotoxic T lymphocytes (CTLs), for example CTL lines, CTL clones, and CTLs from tumor, inflammatory, or other infiltrates.

(79) A “vector” includes plasmids and viruses and any DNA or RNA molecule, whether self-replicating or not, which can be used to transform or transfect a cell.

(80) An “isolated” or “purified” population of cells is substantially free of cells and materials with which it is associated in nature. By substantially free or substantially purified APCs it is meant at least 50% of the population are APCs, preferably at least 70%, more preferably at least 80%, and even more preferably at least 90%, are free of non-APCs cells with which they are associated in nature.

(81) As used herein, a “genetic modification” refers to any addition, deletion or disruption to a cell's normal nucleotides. Any method which can achieve the genetic modification of APCs are within the spirit and scope of this invention. Art recognized methods include viral mediated gene transfer, liposome mediated transfer, transformation, transfection and transduction, e.g., viral-mediated gene transfer such as the use of the improved LAMP Constructs based on DNA viruses such as adenovirus, adeno-associated virus and herpes virus, as well as retroviral based vectors.

(82) The practice of the present invention employs, unless otherwise indicated, conventional molecular biology, microbiology, and recombinant DNA techniques within the skill of the art. Such techniques are explained fully in the literature. See, e.g., Maniatis, Fritsch & Sambrook, In *Molecular Cloning: A Laboratory Manual* (1982); *DNA Cloning: A Practical Approach*, Volumes I and II (D. N. Glover, ed., 1985); *Oligonucleotide Synthesis* (M. J. Gait, ed., 1984); *Nucleic Acid Hybridization* (B. D. Hames & S. J. Higgins, eds., 1985); *Transcription and Translation* (B. D. Hames & S. I. Higgins, eds., 1984); *Animal Cell Culture* (R. I. Freshney, ed., 1986); *Immobilized Cells and Enzymes* (IRL Press, 1986); B. Perbal, *A Practical Guide to Molecular Cloning* (1984).

(83) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. All publications mentioned herein are incorporated by reference for the purpose of describing and disclosing devices, formulations and methodologies that may be used in connection with the presently described invention

(84) LAMP Constructs

(85) LAMP-1, as deduced from a cDNA clone (Chen, et al., *J. Biol. Chem.* 263: 8754, 1988) consists of a polypeptide core of about 382 amino acids with a large (346-residue) luminal amino-terminal domain followed by a 24-residue hydrophobic transmembrane region and short (12-residue) carboxyl-terminal cytoplasmic tail. See, FIGS. 2A and 2B. The luminal domain is highly glycosylated, being substituted with about 20 asparagine linked complex-type oligosaccharides and consists of two approximately 160-residue “homology domains” that are separated by a proline/serine-rich hinge region. Each of these “homology domains” contains 4 uniformly spaced cysteine residues, disulfide bonded to form four 36-38-residue loops symmetrically placed within the two halves of the luminal domain (Arterburn, et al., *J. Biol. Chem.* 265: 7419, 1990; see, also Chen, et al., *J. Biol. Chem.* 25: 263(18): 8754-8, 1988). FIG. 2A schematically shows the conserved domains between LAMP-1, LAMP-2, LAMP-3, Endolyn, LIMBIC, LAMP5, or Macrosailin.

(86) Previously reported LAMP constructs comprise the following elements in this specific

arrangement: (a) a full luminal domain of LAMP-1 protein, the antigen and then the full transmembrane/cytoplasmic tail of LAMP-1 protein; or (b) the antigen and the full transmembrane/cytoplasmic tail of a LAMP-1 protein. In example (a), the antigenic sequence is inserted in between the full luminal domain of a LAMP-1 protein and the LAMP-1 full transmembrane domain/cytoplasmic tail. Both constructs have been shown to successfully target an antigenic sequence to the lysosome/endosome and will be referred to as “complete LAMP Constructs” as shown in FIG. 1 as compared to the improved LAMP Constructs ILC1-ILC6 described herein. The improved LAMP Constructs described herein do not include the complete LAMP Constructs described in the prior art.

(87) Although it has been widely reported in the literature that fragments smaller than the full luminal domain of LAMP-1 were not effective in generating a robust immune response (see, e.g. Godinho et al.). In contrast, the inventors unexpectedly discovered that specific fragments, in certain arrangements, did in fact effectively present allergens to the immune system, generating in many cases a more robust immune response, including the generation of a different repertoire of antibodies. For example, the inventors have identified that the minimal LAMP luminal domain fragment that is effective for generating a robust immune response is not the full luminal domain (as widely reported in the literature) but rather a single Homology Domain of the Luminal Domain of a LAMP Protein.

(88) For example, constructs can comprise, not the full luminal domain, but instead a single Homology Domain of the Luminal Domain of a LAMP Protein. As used herein, the “Homology Domain” comprises at least the 4 uniformly spaced cysteine residues shown in FIGS. 3-10. These cysteine residues are labeled 1, 2, 3, and 4 (and in LIMP-2 and Macrosailin—five cysteines are identified, LIMBIC—six cysteines are identified and Endolyn—eight cysteines are identified) in each Homology Domain as shown in FIGS. 3-10 and are defined herein as the “Cysteine Conserved Fragment.” Additional amino acids can be included to either the N-terminus end and/or the C-terminus end of the Cysteine Conserved Fragment to generate, up to and including a full Homology Domain of a LAMP protein. These additional added amino acids can be derived from the Homology Domain from which the Cysteine Conserved Fragment is derived or from other LAMP Protein Homology Domains. Thus, as used herein, a LAMP Homology Domain comprises and/or consists of one Cysteine Conserved Fragment. At least two LAMP Homology Domains make up the Luminal Domain of LAMP-1, LAMP-2, LAMP-3, or Endolyn.

(89) Specifically, in one preferred embodiment, the improved LAMP Construct comprises at least one allergen described in Table 1/FIG. 14 and/or the combinations described in paragraph [0136-0137] and Table 2 fused to the N-terminus of the luminal domain of a LAMP protein, at least one homology domain of a LAMP protein, or at least one Cysteine Conserved Fragment of a LAMP protein. See, for example ILC-2 and ILC-6 of FIG. 1. In preferred embodiments, these constructs also comprise a transmembrane domain of a LAMP protein, and/or the cytoplasmic tail of a LAMP protein. In other preferred embodiments, when an Allergen X (SEQ ID NO.Y) contains a transmembrane domain, the transmembrane domain of a LAMP protein and/or the cytoplasmic tail of a LAMP protein is unnecessary. In preferred embodiments, two homology domains are included in the improved LAMP Construct (e.g., ILC-1 of FIG. 1). In further preferred embodiments, the two homology domains are derived from a LAMP-1, LAMP-2, LAMP-3, or an Endolyn protein. Alternatively, the two homology domains are derived from different LAMP proteins. In these constructs comprising two homology domains, a LAMP hinge domain may also be included. The improved LAMP Constructs described in this paragraph are unexpected in view of the prior art as the antigen has always been placed in between the full luminal LAMP-1 domain and the full LAMP-1 transmembrane/cytoplasmic tail, as fragments of the luminal domain have not been reported to be effective in generating a robust immune response.

(90) In another preferred embodiment, the improved LAMP Construct comprises at least one allergen described in Table 1/FIG. 14 and/or the combinations described in paragraph [0136-0137]

and Table 2 fused to the C-terminus of a single homology domain of a LAMP protein or a single Cysteine Conserved Fragment of a LAMP protein (e.g., ILC-5 of FIG. 1). In preferred embodiments, these constructs also comprise a transmembrane domain of a LAMP protein, and/or the cytoplasmic tail of a LAMP protein (e.g., ILC-3 of FIG. 1). In other preferred embodiments, when an Allergen X (SEQ ID NO:Y) contains a transmembrane domain, the transmembrane domain of a LAMP protein and/or the cytoplasmic tail of a LAMP protein is unnecessary. Alternatively, two homology domains from two different LAMP proteins may be used. The improved LAMP Constructs described in this paragraph are unexpected in view of the prior art as the allergen has always been placed in between the full luminal LAMP-1 domain and the full LAMP-1 transmembrane/cytoplasmic tail, as fragments of the luminal domain have not been reported to be effective in generating a robust immune response.

(91) Thus, the improved LAMP Construct comprises at least one allergen described in Table 1/FIG. 14 and/or the combinations described in paragraph [0136-0137] and Table 2 fused to the C-terminus of a single homology domain of a LAMP protein or a single Cysteine Conserved Fragment of a LAMP protein. See, for example, ILC-3 and ILC-5 of FIG. 1. In preferred embodiments, these constructs also comprise a transmembrane domain of a LAMP protein, and/or the cytoplasmic tail of a LAMP protein. In other preferred embodiments, when an Allergen X (SEQ ID NO:Y) contains a transmembrane domain, the transmembrane domain of a LAMP protein and/or the cytoplasmic tail of a LAMP protein is unnecessary. The improved LAMP Constructs described in this paragraph are unexpected in view of the prior art as described above.

(92) In another preferred embodiment, the improved LAMP Construct comprises at least one allergen of interest fused in between a first homology domain of a LAMP protein and a second homology domain of a LAMP protein (or at least between two Cysteine Conserved Fragments). See, for example, ILC-4 of FIG. 1. In preferred embodiments, the two homology domains are derived from LAMP-1, LAMP-2, LAMP-3, or an Endolyn protein. In these constructs, the allergen may be placed in the LAMP hinge region. Alternatively, two homology domains from two different LAMP proteins may be used. This arrangement of at least one allergen described in Table 1/FIG. 14 and/or the combinations described in paragraph [0136-0137] and Table 2 fused in between two LAMP homology domains (including Cysteine Conserved Fragments) is unexpected in view of the prior art as described above.

(93) Each of the improved LAMP Constructs defined above can be generated using the domains defined in the Figures. For example, it is specifically contemplated that the domains included in the improved LAMP Construct illustrated in FIG. 1, for example, can originate from sequences derived from orthologous sequences. See, FIGS. 3-10 for example. It is expressly contemplated that the equivalent domains defined in FIGS. 2A and 2B be used to generate the improved LAMP Constructs illustrated in FIG. 1 for orthologous sequences. Moreover, the orthologous sequences shown in FIGS. 3-10 are representative of the sequences that can be used to generate the domains. It is well within the skill in the art to identify other orthologous sequences and/or isotypes and comparing them to the alignments shown in FIGS. 3-10. Thus, by identifying the equivalent boundaries defined in FIGS. 2A and 2B for a human LAMP protein with the alignments shown in FIGS. 3-10, one can generate the improved LAMP Constructs illustrated in FIG. 1.

(94) As would be well understood by the skilled artisan, the boundaries of each domain are an approximation and may be adjusted at least 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acids based on cloning considerations and restriction enzyme placement. Therefore, when a particular domain (e.g., a LAMP Homology Domain) is included in the improved LAMP Construct, the amino acids beginning and ending of the domain may be adjust by at least 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acids as those boundaries defined in FIG. 2A.

(95) Each of the improved LAMP Constructs described above can additionally comprise a signal sequence and/or additional amino acids in between each domain for cloning purposes as is well known in the art. Additionally, the LAMP homologous domains, the LAMP luminal domain, the

LAMP transmembrane domain, and/or the LAMP cytoplasmic tail domain can originate from the same LAMP protein (e.g., human LAMP-1) or different LAMP proteins (e.g., luminal domain from human LAMP-1 and transmembrane domain from human LAMP-2, and/or mixing of orthologous domains in the same gene family (e.g., LAMP-1) or different gene family (LAMP-1 and LAMP-2).

(96) Polypeptide variants of the described LAMP Constructs are contemplated. For example, polypeptides at least about 60%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, 96%, 97%, 98% or 99% identity to any of the improved LAMP Constructs described herein as well as polynucleotides encoding these variants. Variants of the improved LAMP Constructs retain the ability to function by targeting the allergenic sequence to the lysosome. For example, a modified luminal sequence must retain the ability to traffic both membrane and non-membrane antigenic materials to an endosomal compartment with at least about 50%, at least about 60%, at least 70%, at least about 80%, at least about 90%, or at least about 100% efficacy compared to the original domain sequence, i.e., an efficacy that results in sufficient antigen presentation by a cell comprising the chimeric sequence for it to mount an immune response. In one aspect, sequences containing a suitable trafficking signal may be identified by constructing an improved LAMP Construct containing the well-characterized antigenic domain of ovalbumin, a transmembrane domain, and the cytoplasmic domain of a protein containing a putative lysosomal/endosomal targeting signal. Efficiency of targeting can be measured by determining the ability of antigen presenting cells, expressing the improved LAMP Construct, to stimulate HA epitope-specific, MHC class II restricted T-cells (see, e.g., Example 5 of U.S. Pat. No. 5,633,234).

(97) Polynucleotides encoding any of the described improved LAMP Constructs are preferred embodiments of the invention, along with polynucleotides at least about 60%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, 96%, 97%, 98% or 99% identity to any of the improved LAMP Construct polynucleotides described herein. Variants of the improved LAMP Constructs retain the ability to function by targeting the allergenic sequence to the lysosome. For example, a modified luminal sequence must retain the ability to traffic both membrane and non-membrane antigenic materials to an endosomal compartment with at least about 50%, at least about 60%, at least 70%, at least about 80%, at least about 90%, or at least about 100% efficacy compared to the original domain sequence, i.e., an efficacy that results in sufficient antigen presentation by a cell comprising the chimeric sequence for it to mount an immune response. In one aspect, sequences containing a suitable trafficking signal may be identified by constructing an improved LAMP Construct containing the well-characterized antigenic domain of ovalbumin, a transmembrane domain, and the cytoplasmic domain of a protein containing a putative lysosomal/endosomal targeting signal. Efficiency of targeting can be measured by determining the ability of antigen presenting cells, expressing the improved LAMP Construct, to stimulate HA epitope-specific, MHC class II restricted T-cells (see, e.g., Example 5 of U.S. Pat. No. 5,633,234).

(98) Allergens

(99) The following allergens shown in Table 1/FIG. 14 (e.g., Allergen X) can be cloned into each of the LAMP Constructs described herein using techniques well known to the skilled artisan.

Moreover, it is specifically contemplated that any one of the Allergen X (SEQ ID NO:Y) listed in Table 1/FIG. 14 can be combined with any other antigen listed in Table 1/FIG. 14 and inserted into the improved LAMP Constructs as described herein, and specifically as disclosed in Table 2.

(100) As used herein, the term "Allergen X" refers to the specific genes/proteins listed in the following Table 1/FIG. 14, fragments thereof (such as fragments of SEQ ID NO:Y (e.g., as described in column 4 of Table 1) wherein the signal sequence is removed (e.g., as described in column 3 of Table 1)), or mixtures of the listed proteins that are known to induce allergies, i.e., IgE mediated reactions upon their repeated exposure to an individual. Generally, an allergen is any

compound, substance, or material that is capable of evoking an allergic reaction. Allergens are usually understood as a subcategory of antigens, which are compounds, substances, or materials capable of evoking an immune response. For carrying out the invention, Allergen X may be selected, among other things, from natural or native allergens, modified natural allergens, synthetic allergens, recombinant allergens, allergoids, and mixtures or combinations thereof. Of particular interest is Allergen X that is capable of causing an IgE-mediated immediate type hypersensitivity. (101) As used herein, the amino acid sequence of Allergen X comprises any one of SEQ ID NO:Y (with or without the signal sequence). Representative examples of polynucleotides that can encode Allergen X (SEQ ID NO:Y) are shown as SEQ ID NO:Z in Table 1/FIG. 14 or any polynucleotide (such as a codon optimized sequence) that encodes the Allergen X as described in column 2 or the fragments described in column 4 of Table 1. These polynucleotides are inserted into any one of the Constructs ILC1-6 as shown in FIG. 1 (with or without the signal sequence of SEQ ID NO:Y). Insertion may be facilitated by the use of cloning sequences that flank the amino acid sequence of SEQ ID NO:Y, such as, for example, polynucleotides that encode “Leu-Glu” and “Glu-Phe” (e.g., “CTCGAG” and “GAATTC.”) As used herein, “Allergen X” also encompasses variants (including fragments) of Allergen X. For example, preferred embodiments include Allergen X polypeptide variants that have at least about 60%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, 96%, 97%, 98% or 99% identity to SEQ ID NO: Y (as described in column 2 or column 4 of Table 1). These variants retain either (a) the ability to raise an antibody that cross-reacts with the Allergen X from which it was derived and/or (b) Allergen X biological activity. Polynucleotides that encode these variant Allergen X polypeptides are specifically contemplated.

(102) The invention further provides a nucleic acid molecule encoding any of the Allergen X of Table 1/FIG. 14. The invention also provides a vector comprising the nucleic acid encoding Allergen X (SEQ ID NO:Y), such as, for example, SEQ ID NO:Z or any polynucleotide (such as a codon optimized sequence) that encodes the Allergen X as described in column 2 or the fragments described in column 4 of Table 1, wherein the nucleic acid molecule is operably linked to an expression control sequence in any one of ILC1-6. In one preferred aspect, the vector is a vaccine vector, suitable for vaccinating a patient against Allergen X. In another aspect, the invention provides a delivery vehicle comprising the nucleic acid molecule for facilitating the introduction of the nucleic acid molecule into a cell. The delivery vehicle may be lipid-based (e.g., a liposome formulation), viral-based (e.g., comprising viral proteins encapsulating the nucleic acid molecule), or cell-based. In one preferred aspect, the vector is a vaccine vector. As would be well understood by the skilled artisan, columns 1 and 4 of Table 1 defines the preferred amino acids to be cloned into the improved LAMP Constructs described herein. However, both the N-terminal and the C-terminal boundaries of the fragments described in columns 1 and 4 of Table 1 are an approximation and may be adjusted at least by 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acids based on cloning considerations and restriction enzyme placement. Therefore, when Allergen X is included in the improved LAMP Construct, the amino acids beginning and ending of Allergen X may be adjust by at least 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acids as defined in columns 1 and 4 of Table 1.

(103) TABLE-US-00001 Amino Acids of Representative Amino SEQ ID NO: Y Preferred Amino Acids Polynucleotide Acid Sequence relating to of SEQ ID NO: Y to be Sequences Encoding of Allergen X Signal included in improved SEQ ID NO: Y Allergen X (SEQ ID NO: Y) Sequence LAMP Constructs (SEQ ID NO: Z)

Cor a 1	SEQ ID NO: 115	SEQ ID NO: 115	SEQ ID NO: 114	Cor a 9
SEQ ID NO: 117	SEQ ID NO: 117	SEQ ID NO: 116	Cor a 1 - Gly4 -	
Cor a 9	SEQ ID NO: 119	SEQ ID NO: 119	SEQ ID NO: 118	Pru du 6
SEQ ID NO: 121	SEQ ID NO: 121	SEQ ID NO: 120	Ana o 1	SEQ ID NO: 123
SEQ ID NO: 123	SEQ ID NO: 123	SEQ ID NO: 122	Ana o 2	SEQ ID NO: 125
SEQ ID NO: 125	SEQ ID NO: 124	Ana o 3	SEQ ID NO: 127	SEQ ID NO: 125

NO: 127 SEQ ID NO: 126 Ana 0 2 - Gly4 - Ana 0 1 - SEQ ID
 NO: 129 SEQ ID NO: 129 SEQ ID NO: 128 Gly4 - Ana 0 3 Jug n 1
 SEQ ID NO: 131 SEQ ID NO: 131 SEQ ID NO: 130 Jug r 2 SEQ ID
 NO: 133 SEQ ID NO: 133 SEQ ID NO: 132 Jug n 1 - Gly4 - Jug r 2
 SEQ ID NO: 135 SEQ ID NO: 135 SEQ ID NO: 134 Amb a 1 SEQ ID
 NO: 137 SEQ ID NO: 137 Bet v 1-A SEQ ID NO: 141 SEQ ID NO: 141
 Can f 1 SEQ ID NO: 145 SEQ ID NO: 145 Cyn d 1 SEQ ID NO: 149
 SEQ ID NO: 149 Der F 1 (19-321) SEQ ID NO: 153 SEQ ID NO: 153
 Der F 1 (99-321) SEQ ID NO: 157 SEQ ID NO: 157 Der P2 SEQ ID NO:
 161 SEQ ID NO: 161 Der F 2 SEQ ID NO: 165 SEQ ID NO: 165 DerP1
 (del) SEQ ID NO: 169 SEQ ID NO: 169 Fel D 1 SEQ ID NO: 173 SEQ
 ID NO: 173 SEQ ID NO: 174 SEQ ID NO: 174 Fel d 2 SEQ ID NO: 178
 SEQ ID NO: 178 Fel d 4 SEQ ID NO: 182 SEQ ID NO: 182 Lit v 1
 SEQ ID NO: 186 SEQ ID NO: 186 Lol p 5a SEQ ID NO: 190 SEQ ID
 NO: 190 Phl p 1 SEQ ID NO: 194 SEQ ID NO: 194 Phl p 5 SEQ ID
 NO: 198 SEQ ID NO: 198 Der f 15 SEQ ID NO: 201 SEQ ID NO: 201
 Derf 18 SEQ ID NO: 202 SEQ ID NO: 202 Zen-1 SEQ ID NO: 203 SEQ ID
 NO: 203 Cte f 1 SEQ ID NO: 204 SEQ ID NO: 204 Der F15-Der F18 SEQ
 ID NO: 205 SEQ ID NO: 205 Der F1-Der F2 SEQ ID NO: 206 SEQ ID
 NO: 206 Api m1 SEQ ID NO: 207 SEQ ID NO: 207 Api m2 SEQ ID NO:
 208 SEQ ID NO: 208 Api m3 SEQ ID NO: 209 SEQ ID NO: 209 Api m5
 SEQ ID NO: 210 SEQ ID NO: 210 Api m10 SEQ ID NO: 211 SEQ ID
 NO: 211 Ves v1 SEQ ID NO: 212 SEQ ID NO: 212 Ves v2 SEQ ID NO: 213
 SEQ ID NO: 213 Ves v3 SEQ ID NO: 214 SEQ ID NO: 214 Ves v5 SEQ
 ID NO: 215 SEQ ID NO: 215 Pol d1 SEQ ID NO: 216 SEQ ID NO: 216
 Pol d5 SEQ ID NO: 217 SEQ ID NO: 217 Ara H1 SEQ ID NO: 218 1-25 26-
 626 Ara H2 SEQ ID NO: 219 SEQ ID NO: 219 Ara H3 SEQ ID NO: 220 1-20
 21-512 AraH1-AraH2-AraH3 SEQ ID NO: 221 SEQ ID NO: 221 Cry J1 (*C.*
japonica) SEQ ID NO: 222 1-21 22-374 Cry J2 (*C. japonica*) SEQ ID NO: 223 1-
 22 23-514 CryJ1-Cry J2 (*C. japonica*) SEQ ID NO: 224 SEQ ID NO: 224 Cry
 J3 (Cry 13.8; *C. japonica*) SEQ ID NO: 225 1-26 27-225 CJP-4 (*C. japonica*)
 SEQ ID NO: 226 1-34 35-281 CJP-6 (*C. japonica*) SEQ ID NO: 227 1-35 36-306
 CJP-8 (*C. japonica*) SEQ ID NO: 228 1-35 36-165 CPA63 (*C. japonica*) SEQ ID
 NO: 229 1-20 21-472 CJP38 (*C. japonica*) SEQ ID NO: 230 1-28 29-348 Cha o 1
 (*C. obtuse*) SEQ ID NO: 231 1-21 22-375 Jun a 1 (*J. ashei*) SEQ ID NO: 232
 1-21 22-367 Jun v 1 (*J. virginiana*) SEQ ID NO: 233 1-21 22-367 Cup a 1 (*H.*
arizonica) SEQ ID NO: 234 1-21 22-367 Jun o 1 (*J. oxycedrus*) SEQ ID NO:
 235 1-21 22-366 Cup s 1 (*C. sempervirens*) SEQ ID NO: 236 1-21 22-367 Cha o
 2 (*C. obtuse*) (signal SEQ ID NO: 237 1-50 51-514 sequence is residues 1-22)
 Jun a 2 (*J. ashei*) (signal SEQ ID NO: 238 1-53 54-507 sequence is
 residues 1-22) Cup a 2 (*H. arizonica*) SEQ ID NO: 239 SEQ ID NO: 239
 Jun a 3 (*J. ashei*) (signal SEQ ID NO: 240 1-26 27-225 sequence is
 residues 1-26) Jun r 3 (*J. rigida*) SEQ ID NO: 241 1-26 27-225 Cup s 3 (*C.*
sempervirens) SEQ ID NO: 242 1-26 27-224 Cup a 3 (*H. arizonica*) SEQ ID
 NO: 243 SEQ ID NO: 243 Ch4A (*P. monticola*) SEQ ID NO: 244 1-25 26-274
 Ch4-1 (*P. menziesii*; SEQ ID NO: 245 1-26 27-277 SEQ ID NO: 30) PT-1 (*P.*
taeda) SEQ ID NO: 246 SEQ ID NO: 246 LTP (*P. abies*) SEQ ID NO: 247 1-
 25 26-173 Ara H 6 SEQ ID NO: 248 SEQ ID NO: 248 Ara H 8 SEQ ID
 NO: 249 SEQ ID NO: 249 Ara H 9 SEQ ID NO: 250 SEQ ID NO: 250
 (104) Additionally, more than one Allergen X can be combined (in any order) and administered as a

vaccine as in any one of the improved LAMP Constructs as described herein. It is specifically envisioned that the combinations of Allergen X can be cloned within a single improved LAMP construct, or can be delivered in a composition comprising multiple improved LAMP constructs of Allergen X. Specifically, Allergen X as described in Table 1/FIG. 14 can be cloned into the improved LAMP Constructs described herein either individually, or in combination with one another Allergen X as described in Table 2. The order of the combination of Allergen X as described in Table 2 for a particular improved LAMP construct can vary as this list describes what an improved LAMP Construct comprises and not necessarily to describe the arrangement of the Allergen X within a particular construct.

(105) In preferred embodiments, the improved LAMP Constructs described herein comprise, or consist, (a) any one of the polynucleotides of SEQ ID NO:Z as shown in Table 1/FIG. 14; (b) a polynucleotide encoding a polypeptide encoded by any one of the polynucleotides of SEQ ID NO:Z as shown in Table 1/FIG. 14; (c) a polynucleotide that encodes a polypeptide comprising the amino acid sequence as described in either column 2 or column 4 of SEQ ID NO: Y as shown in Table 1/FIG. 14; or (g) polynucleotides with at least about 60%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, 96%, 97%, 98% or 99% identity to any one of the polynucleotides of (a)-(c). Polypeptides encoded by these polynucleotides are additional preferred embodiments.

(106) TABLE-US-00002 TABLE 2 Allergen X Can be combined with any one or more of the following Allergens: Cor a 1 Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cor a 9 Cor a 1, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Pru du 6 Cor a 1, Cor a 9, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Ana o 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet

v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Ana o 2 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Ana o 3 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Jug n 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2,+10 Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Jug r 2 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Amb a 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel

d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Bet v 1-A Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Can f1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cyn d 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Der F 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (99-321), Der P2, Der F 2, DerP1 (19-321) (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Der F 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der P2, Der F 2, DerP1 (99-321) (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1,

Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5,
 Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3,
 CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a
 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r
 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8,
 and/or Ara H 9 Der P2 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana
 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1,
 Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der F 2, DerP1 (del),
 Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5,
 Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api
 m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara
 H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8,
 CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups
 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup
 a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Der
 F 2 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3,
 Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F
 1 (19-321), Der F 1 (99-321), Der P2, DerP1 (del), Fel d 1, Fel d 2,
 Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f
 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves
 v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara
 H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o
 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a
 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-
 1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 DerP1 (del) Cor a 1, Cor
 a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2,
 Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1
 (99-321), Der P2, Der F 2, Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol
 p 5a, Phl p 1, Phl p 5, Der f 15, Derf 18, Zen-1, Cte f 1, Api
 m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves
 v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry
 J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1,
 Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3,
 Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara
 H 8, and/or Ara H 9 Fel D 1 Cor a 1, Cor a 9, Pru du 6, Ana 0
 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A,
 Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F
 2, DerP1 (del), Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl
 p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3,
 Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5,
 Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-
 8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1,
 Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3,
 Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9
 Fel d 2 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3,
 Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F
 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1,
 Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f
 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves
 v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara

H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o
1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a
2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-
1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Fel d 4 Cor a 1, Cor a
9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb
a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-
321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Lit v 1, Lol
p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api
m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves
v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry
J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1,
Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a
3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6,
Ara H 8, and/or Ara H 9 Lit v 1 Cor a 1, Cor a 9, Pru du 6, Ana
0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A,
Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der
F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lol p 5a, Phl p 1,
Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api
m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol
d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6,
CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1,
Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s
3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara
H 9 Lol p 5a Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana
0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1,
Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel
d 1, Fel d 2, Fel d 4, Lit v 1, Phl p 1, Phl p 5, Der f 15, Der
f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves
v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara
H3, Cry J1, Cry J2,+10 Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha
o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun
a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1,
PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Phl p 1 Cor a 1, Cor
a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2,
Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F
1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4,
Lit v 1, Lol p 5a, Phl p 5, Derf 15, Der f 18, Zen-1, Cte f 1,
Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3,
Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2,
Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v
1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a
3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6,
Ara H 8, and/or Ara H 9 Phl p 5 Cor a 1, Cor a 9, Pru du 6, Ana
0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A,
Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der
F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a,
Phl p 1, Derf 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api
m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol
d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6,
CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1,

Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3,
 Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9
 Der f 15 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0
 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1,
 Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel
 d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5,
 Derf 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10,
 Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara
 H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o
 1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a
 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-
 1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Der f 18 Cor a 1, Cor a
 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb
 a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-
 321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v
 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Zen-1, Cte f 1, Api m1,
 Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5,
 Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3,
 CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a
 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r
 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8,
 and/or Ara H 9 Zen-1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0
 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn
 d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1
 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl
 p 5, Der f 15, Der f 18, Cte f 1, Api m1, Api m2, Api m3, Api
 m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara
 H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8,
 CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cups
 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup
 a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cte
 f 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug
 n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1
 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Feld 1, Fel d
 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Derf
 18, Zen-1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves
 v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry
 J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun
 a 1, Jun v1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup
 a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP,
 Ara H 6, Ara H 8, and/or Ara H 9 Api m1 Cor a 1, Cor a 9, Pru
 du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1,
 Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321),
 Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1,
 Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1,
 Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5,
 Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3,
 CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a
 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r
 3, Cup s+10 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8,

and/or Ara H 9 Api m2 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Api m3 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Api m5 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s+10 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Api m10 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1,+10 Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Ves v1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Ves v2 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r

2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Ves v3 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry+10 J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Ves v5 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v3, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Pol d1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Pol d5 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Ara H1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1

(del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Ara H2 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Ara H3 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cry J1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup+10 s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cry J2 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cry J3 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-

1, Cte f 1, Api m1, Api m2, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 CJP-4 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 CJP-6 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 CJP-8 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 CPA63 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 CJP38 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl+10 p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5,

Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cha o 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Jun a 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, +10 Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Jun v 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Cup a 1, Jun o 1, Cups 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cup a 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Jun o 1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38,

Cha o 1, Jun a1, Jun v 1, Cup a 1, Cup s 1, Cha o 2, Jun a 2,
 Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1,
 LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cup s 1 Cor a 1, Cor a 9,
 Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a
 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-
 321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v
 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f
 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves
 v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry
 J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a1, Jun
 v 1, Cup a 1, Jun o 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3,
 Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara
 H 8, and/or Ara H 9 Cha o 2 Cor a 1, Cor a 9, Pru du 6, Ana 0
 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A,
 Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der
 F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a,
 Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1,
 Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v3, Ves v5,
 Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3,
 CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a1, Jun v 1, Cup a
 1, Jun o 1, Cup s 1, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s
 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara
 H 9 Jun a 2 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana
 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1,
 Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel
 d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der
 f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5,
 Apim10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1,
 Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63,
 CJP38, Cha o 1, Jun a1, Jun v 1, Cup a 1, Jun o 1, Cups 1, Cha
 o 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1,
 PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cup a 2 Cor a 1, Cor
 a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2,
 Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F
 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4,
 Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1,
 Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2,
 Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1,
 Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a1,
 Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Jun a
 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6,
 Ara H 8, and/or Ara H 9 Jun a 3 Cor a 1, Cor a 9, Pru du 6, Ana
 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A,
 Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der
 F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a,
 Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1,
 Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v3, Ves v5,
 Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3,
 CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a1, Jun v 1, Cup a
 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun r 3, Cup

s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or
 Ara H 9 Jun r 3 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2,
 Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn
 d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1
 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl
 p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3,
 Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5,
 Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-
 8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1,
 Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Cup s 3, Cup a
 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cup s
 3 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n
 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-
 321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d
 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der
 f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api m10,
 Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2,
 Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38,
 Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2,
 Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup a 3, Ch4A, Ch4-1, PT-1,
 LTP, Ara H 6, Ara H 8, and/or Ara H 9 Cup a 3 Cor a 1, Cor a 9,
 Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a
 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-
 321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v
 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f
 1, Api m1, Api m2, Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves
 v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry
 J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun
 v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2,
 Jun a 3, Jun r 3, Cup s 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, Ara
 H 8, and/or Ara H 9 Ch4A Cor a 1, Cor a 9, Pru du 6, Ana 0 1,
 Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can
 f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2,
 DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p
 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2,
 Api m3, Api m5, Api m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1,
 Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-
 6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o
 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup
 s 3, Cup a 3, Ch4-1, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9
 Ch4-1 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3,
 Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F
 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1,
 Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f
 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Api
 m10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara
 H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63,
 CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1,
 Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a
 3, Ch4A, PT-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 PT-1 Cor a 1,

Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, LTP, Ara H 6, Ara H 8, and/or Ara H 9 LTP Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, Ara H 6, Ara H 8, and/or Ara H 9 Ara H 6 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Can f 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 8, and/or Ara H 9 Ara H 8 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, and/or Ara H 9 Ara H 9 Cor a 1, Cor a 9, Pru du 6, Ana 0 1, Ana 0 2, Ana 0 3, Jug n 1, Jug r 2, Amb a 1, Bet v 1-A, Canf 1, Cyn d 1, Der F 1 (19-321), Der F 1 (99-321), Der P2, Der F 2, DerP1 (del), Fel d 1, Fel d 2, Fel d 4, Lit v 1, Lol p 5a, Phl p 1, Phl p 5, Der f 15, Der f 18, Zen-1, Cte f 1, Api m1, Api m2, Api m3, Api m5, Apim10, Ves v1, Ves v2, Ves v3, Ves v5, Pol d1, Pol d5, Ara H1, Ara H2, Ara H3, Cry J1, Cry J2, Cry J3, CJP-4, CJP-6, CJP-8, CPA63, CJP38, Cha o 1, Jun a 1, Jun v 1, Cup a 1, Jun o 1, Cup s 1, Cha o 2, Jun a 2, Cup a 2, Jun a 3, Jun r 3, Cup s 3, Cup a 3, Ch4A, Ch4-1, PT-1, LTP, Ara H 6, and/or Ara H 8

(107) Procedures for constructing improved LAMP Constructs comprising the allergen of interest are well known in the art (see e.g., Williams, et al., J. Cell Biol. 111: 955, 1990). DNA sequences encoding the desired segments can be obtained from readily available recombinant DNA materials such as those available from the American Type Culture Collection, 12301 Parklawn Drive, Rockville, Md. 20852, U.S.A., or from DNA libraries that contain the desired DNA.

(108) For example, the DNA segments corresponding to the desired domain sequences can be assembled with appropriate control and signal sequences using routine procedures of recombinant DNA methodology. See, e.g., as described in U.S. Pat. No. 4,593,002, and Langford, et al., Molec. Cell. Biol. 6: 3191, 1986.

(109) A DNA sequence encoding a protein or polypeptide can be synthesized chemically or isolated by one of several approaches. The DNA sequence to be synthesized can be designed with the appropriate codons for the desired amino acid sequence. In general, one will select preferred codons for the intended host in which the sequence will be used for expression. The complete sequence may be assembled from overlapping oligonucleotides prepared by standard methods and assembled into a complete coding sequence. See, e.g., Edge, Nature 292: 756, 1981; Nambair, et al. Science 223: 1299, 1984; Jay, et al., J. Biol. Chem. 259: 6311, 1984.

(110) In one aspect, one or more of the nucleic acids encoding the domain sequences of the improved LAMP Construct are isolated individually using the polymerase chain reaction (M. A. Innis, et al., In PCR Protocols: A Guide to Methods and Applications, Academic Press, 1990). The domains are preferably isolated from publicly available clones known to contain them, but they may also be isolated from genomic DNA or cDNA libraries. Preferably, isolated fragments are bordered by compatible restriction endonuclease sites which allow an improved LAMP Construct encoding the allergen sequence to be constructed. This technique is well known to those of skill in the art. Domain sequences may be fused directly to each other (e.g., with no intervening sequences), or inserted into one another (e.g., where domain sequences are discontinuous), or may be separated by intervening sequences (e.g., such as linker sequences).

(111) The basic strategies for preparing oligonucleotide primers, probes and DNA libraries, as well as their screening by nucleic acid hybridization, are well known to those of ordinary skill in the art. See, e.g., Sambrook, et al., 1989, supra; Perbal, 1984, supra. The construction of an appropriate genomic DNA or cDNA library is within the skill of the art. See, e.g., Perbal, 1984, supra. Alternatively, suitable DNA libraries or publicly available clones are available from suppliers of biological research materials, such as Clontech and Stratagene, as well as from public depositories such as the American Type Culture Collection.

(112) Selection may be accomplished by expressing sequences from an expression library of DNA and detecting the expressed peptides immunologically. Clones which express peptides that bind to MHC II molecules and to the desired antibodies/T cell receptors are selected. These selection procedures are well known to those of ordinary skill in the art (see, e.g., Sambrook, et al., 1989, supra).

(113) Once a clone containing the coding sequence for the desired polypeptide sequence has been prepared or isolated, the sequence can be cloned into any suitable vector, preferably comprising an origin of replication for maintaining the sequence in a host cell.

(114) Nucleic Acid Delivery Vehicles

(115) In one aspect, a vaccine composition comprising an improved LAMP Construct is introduced into a cell. The cell may be a host cell for replicating the nucleic acid or for expressing the improved LAMP Construct. Preferably, the host cell for expressing the improved LAMP Construct is an antigen presenting cell (described further below).

(116) In preferred embodiments, the improved LAMP Construct further comprises a polynucleotide sequence for insertion into a target cell and an expression control sequence operably linked thereto to control expression of the polynucleotide sequence (e.g., transcription and/or translation) in the cell. Examples include plasmids, phages, autonomously replicating sequences (ARS), centromeres,

and other sequences which are able to replicate or be replicated in vitro or in a host cell (e.g., such as a bacterial, yeast, or insect cell) and/or target cell (e.g., such as a mammalian cell, preferably an antigen presenting cell) and/or to convey the sequences encoding the improved LAMP Construct to a desired location within the target cell.

(117) Recombinant expression vectors may be derived from micro-organisms which readily infect animals, including man, horses, cows, pigs, llamas, giraffes, dogs, cats or chickens. Preferred vectors include those which have already been used as live vaccines, such as vaccinia. These recombinants can be directly inoculated into a host, conferring immunity not only to the microbial vector, but also to express foreign allergens. Preferred vectors contemplated herein as live recombinant vaccines include RNA viruses, adenovirus, herpesviruses, poliovirus, and vaccinia and other pox viruses, as taught in Flexner, Adv. Pharmacol. 21: 51, 1990, for example.

(118) Expression control sequences include, but are not limited to, promoter sequences to bind RNA polymerase, enhancer sequences or negative regulatory elements to bind to transcriptional activators and repressors, respectively, and/or translation initiation sequences for ribosome binding. For example, a bacterial expression vector can include a promoter such as the lac promoter and for transcription initiation, the Shine-Dalgarno sequence and the start codon AUG (Sambrook, et al., 1989, supra). Similarly, a eukaryotic expression vector preferably includes a heterologous, homologous, or chimeric promoter for RNA polymerase II, a downstream polyadenylation signal, the start codon AUG, and a termination codon for detachment of a ribosome.

(119) Expression control sequences may be obtained from naturally occurring genes or may be designed. Designed expression control sequences include, but are not limited to, mutated and/or chimeric expression control sequences or synthetic or cloned consensus sequences. Vectors that contain both a promoter and a cloning site into which a polynucleotide can be operatively linked are well known in the art. Such vectors are capable of transcribing RNA in vitro or in vivo, and are commercially available from sources such as Stratagene (La Jolla, Calif.) and Promega Biotech (Madison, Wis.).

(120) In order to optimize expression and/or transcription, it may be necessary to remove, add or alter 5' and/or 3' untranslated portions of the vectors to eliminate extra, or alternative translation initiation codons or other sequences that may interfere with, or reduce, expression, either at the level of transcription or translation. Alternatively, consensus ribosome binding sites can be inserted immediately 5' of the start codon to enhance expression. A wide variety of expression control sequences—sequences that control the expression of a DNA sequence operatively linked to it—may be used in these vectors to express the DNA sequences of this invention. Such useful expression control sequences include, for example, the early or late promoters of SV40, CMV, vaccinia, polyoma, adenovirus, herpes virus and other sequences known to control the expression of genes of mammalian cells, and various combinations thereof.

(121) In one aspect, the improved LAMP Construct comprises an origin of replication for replicating the vector. Preferably, the origin functions in at least one type of host cell which can be used to generate sufficient numbers of copies of the sequence for use in delivery to a target cell. Suitable origins therefore include, but are not limited to, those which function in bacterial cells (e.g., such as *Escherichia* sp., *Salmonella* sp., *Proteus* sp., *Clostridium* sp., *Klebsiella* sp., *Bacillus* sp., *Streptomyces* sp., and *Pseudomonas* sp.), yeast (e.g., such as *Saccharomyces* sp. or *Pichia* sp.), insect cells, and mammalian cells. In one preferred aspect, an origin of replication is provided which functions in the target cell into which the nucleic acid delivery vehicle is introduced (e.g., a mammalian cell, such as a human cell). In another aspect, at least two origins of replication are provided, one that functions in a host cell and one that functions in a target cell.

(122) The improved LAMP Construct may alternatively, or additionally, comprise sequences to facilitate integration of at least a portion of the nucleic acid deliver vector into a target cell chromosome. For example, the improved LAMP Construct may comprise regions of homology to target cell chromosomal DNA. In one aspect, the delivery vector comprises two or more

recombination sites which flank a nucleic acid sequence encoding the improved LAMP Construct. (123) The vector may additionally comprise a detectable and/or selectable marker to verify that the vector has been successfully introduced in a target cell and/or can be expressed by the target cell. These markers can encode an activity, such as, but not limited to, production of RNA, peptide, or protein, or can provide a binding site for RNA, peptides, proteins, inorganic and organic compounds or compositions and the like.

(124) Examples of detectable/selectable markers genes include, but are not limited to: DNA segments that encode products which provide resistance against otherwise toxic compounds (e.g., antibiotics); DNA segments that encode products which are otherwise lacking in the recipient cell (e.g., tRNA genes, auxotrophic markers); DNA segments that encode products which suppress the activity of a gene product; DNA segments that encode products which can be readily identified (e.g., phenotypic markers such as beta-galactosidase, a fluorescent protein (GFP, CFP, YFP, BFP, RFP, EGFP, EYFP, EBFP, dsRed, mutated, modified, or enhanced forms thereof, and the like), and cell surface proteins); DNA segments that bind products which are otherwise detrimental to cell survival and/or function; DNA segments that otherwise inhibit the activity of other nucleic acid segments (e.g., antisense oligonucleotides); DNA segments that bind products that modify a substrate (e.g., restriction endonucleases); DNA segments that can be used to isolate or identify a desired molecule (e.g., segments encoding specific protein binding sites); primer sequences; DNA segments, which when absent, directly or indirectly confer resistance or sensitivity to particular compounds; and/or DNA segments that encode products which are toxic in recipient cells.

(125) The marker gene can be used as a marker for confirmation of successful gene transfer and/or to isolate cells expressing transferred genes and/or to recover transferred genes from a cell. For example, in one aspect, the marker gene is used to isolate and purify antigen presenting cells expressing the improved LAMP Constructs.

(126) Substantially similar genes maybe provided, e.g., genes with greater than about 50%, greater than about 70%, greater than 80%, greater than about 90%, and preferably, greater than about 95% identity to a known gene. Substantially similar domain sequences may initially be identified by selecting a sequence which specifically hybridizes to a domain sequence of interest under stringent hybridization conditions. Performing assays to determine the suitability of homologous, variant, or modified domain sequences is merely a matter of screening for sequences which express the appropriate activity. Such screening is routine in the art.

(127) The improved LAMP Construct may be provided as naked nucleic acids or in a delivery vehicle associated with one or more molecules for facilitating entry of a nucleic acid into a cell. Suitable delivery vehicles include, but are not limited to: liposomal formulations, polypeptides, polysaccharides, lipopolysaccharides, viral formulations (e.g., including viruses, viral particles, artificial viral envelopes and the like), cell delivery vehicles, and the like.

(128) Lipid-Based Formulations

(129) Delivery vehicles designed to facilitate intracellular delivery of the improved LAMP Constructs must interact with both non-polar and polar environments (in or on, for example, the plasma membrane, tissue fluids, compartments within the cell, and the like). Therefore, preferably, delivery vehicles are designed to contain both polar and non-polar domains or a translocating sequence for translocating an improved LAMP Construct into a cell.

(130) Compounds having polar and non-polar domains are termed amphiphiles. Cationic amphiphiles have polar groups that are capable of being positively charged at, or around, physiological pH for interacting with negatively charged polynucleotides such as DNA.

(131) The improved LAMP Constructs described herein can be provided in formulations comprising lipid monolayers or bilayers to facilitate transfer of the vectors across a cell membrane. Liposomes or any form of lipid membrane, such as planar lipid membranes or the cell membrane of an intact cell, e.g., a red blood cell, can be used. Liposomal formulations can be administered by any means, including administration intravenously or orally.

(132) Liposomes and liposomal formulations can be prepared according to standard methods and are well known in the art, see, e.g., Remington's; Akimaru, 1995, *Cytokines Mol. Ther.* 1: 197-210; Alving, 1995, *Immunol. Rev.* 145: 5-31; Szoka, 1980, *Ann. Rev. Biophys. Bioeng.* 9:467; U.S. Pat. Nos. 4,235,871; 4,501,728; and 4,837,028. In one aspect, the liposome comprises a targeting molecule for targeting a liposome:improved LAMP Construct complex to a particular cell type. In a particularly preferred aspect, a targeting molecule comprises a binding partner (e.g., a ligand or receptor) for a biomolecule (e.g., a receptor or ligand) on the surface of a blood vessel or a cell found in a target tissue.

(133) Liposome charge is an important determinant in liposome clearance from the blood, with negatively charged liposomes being taken up more rapidly by the reticuloendothelial system (Juliano, 1975, *Biochem. Biophys. Res. Commun.* 63: 651) and thus having shorter half-lives in the bloodstream. Incorporating phosphatidylethanolamine derivatives enhances the circulation time by preventing liposomal aggregation. For example, incorporation of N-(omega-carboxy)acylamidophosphatidylethanolamines into large unilamellar vesicles of L-alpha-distearoylphosphatidylcholine dramatically increases the in vivo liposomal circulation lifetime (see, e.g., Ahl, 1997, *Biochim. Biophys. Acta* 1329: 370-382). Liposomes with prolonged circulation half-lives are typically desirable for therapeutic and diagnostic uses. For a general discussion of pharmacokinetics, see, e.g., Remington's, Chapters 37-39, Lee, et al., In *Pharmacokinetic Analysis: A Practical Approach* (Technomic Publishing AG, Basel, Switzerland 1996).

(134) Typically, liposomes are prepared with about 5 to 15 mole percent negatively charged phospholipids, such as phosphatidylglycerol, phosphatidylserine or phosphatidyl-inositol. Added negatively charged phospholipids, such as phosphatidylglycerol, also serve to prevent spontaneous liposome aggregation, and thus minimize the risk of undersized liposomal aggregate formation. Membrane-rigidifying agents, such as sphingomyelin or a saturated neutral phospholipid, at a concentration of at least about 50 mole percent, and 5 to 15 mole percent of monosialylganglioside can also impart desirably liposome properties, such as rigidity (see, e.g., U.S. Pat. No. 4,837,028). (135) Additionally, the liposome suspension can include lipid-protective agents which protect lipids against free-radical and lipid-peroxidative damages on storage. Lipophilic free-radical quenchers, such as alpha-tocopherol and water-soluble iron-specific chelators, such as ferrioxianine, are preferred.

(136) The improved LAMP Constructs of the invention can include multilamellar vesicles of heterogeneous sizes. For example, vesicle-forming lipids can be dissolved in a suitable organic solvent or solvent system and dried under vacuum or an inert gas to form a thin lipid film. If desired, the film can be redissolved in a suitable solvent, such as tertiary butanol, and then lyophilized to form a more homogeneous lipid mixture which is in a more easily hydrated powder like form. This film is covered with an aqueous solution of the peptide or polypeptide complex and allowed to hydrate, typically over a 15 to 60 minute period with agitation. The size distribution of the resulting multilamellar vesicles can be shifted toward smaller sizes by hydrating the lipids under more vigorous agitation conditions or by adding solubilizing detergents such as deoxycholate. The hydration medium preferably comprises the nucleic acid at a concentration which is desired in the interior volume of the liposomes in the final liposome suspension.

(137) Following liposome preparation, the liposomes can be sized to achieve a desired size range and relatively narrow distribution of liposome sizes. One preferred size range is about 0.2 to 0.4 microns, which allows the liposome suspension to be sterilized by filtration through a conventional filter, typically a 0.22 micron filter. Filter sterilization can be carried out on a high throughput basis if the liposomes have been sized down to about 0.2 to 0.4 microns. Several techniques are available for sizing liposome to a desired size (see, e.g., U.S. Pat. No. 4,737,323).

(138) Suitable lipids include, but are not limited to, DOTMA (Felgner, et al., 1987, *Proc. Natl. Acad. Sci. USA* 84: 7413-7417), DOGS or Transfectain™ (Behr, et al., 1989, *Proc. Natl. Acad. Sci. USA* 86: 6982-6986), DNERIE or DORIE (Felgner, et al., *Methods* 5: 67-75), DC-CHOL (Gao and

Huang, 1991, BBRC 179: 280-285), DOTAP™ (McLachlan, et al., 1995, Gene Therapy 2: 674-622), Lipofectamine™. and glycerolipid compounds (see, e.g., EP901463 and WO98/37916).

(139) Other molecules suitable for complexing with the improved LAMP Constructs include cationic molecules, such as, polyamidoamine (Haensler and Szoka, 1993, Bioconjugate Chem. 4: 372-379), dendritic polysine (WO 95/24221), polyethylene irinine or polypropylene h-nine (WO 96/02655), polylysine (U.S. Pat. No. 5,595,897; FR 2 719 316), chitosan (U.S. Pat. No. 5,744,166), DNA-gelatin coarccervates (see, e.g., U.S. Pat. Nos. 6,207,195; 6,025,337; 5,972,707) or DEAE dextran (Lopata, et al., 1984, Nucleic Acid Res. 12: 5707-5717).

(140) Viral-Based Gene Delivery Vehicles

(141) In one aspect, the improved LAMP Construct delivery vehicle comprises a virus or viral particle. In this aspect, preferably, the improved LAMP Construct comprises a viral vector. Viral vectors, such as retroviruses, adenoviruses, adeno-associated viruses and herpes viruses, are often made up of two components, a modified viral genome and a coat structure surrounding it (see, e.g., Smith et al., 1995, Ann. Rev. Microbiol. 49: 807-838), although sometimes viral vectors are introduced in naked form or coated with proteins other than viral proteins. Most current vectors have coat structures similar to a wild-type virus. This structure packages and protects the viral nucleic acid and provides the means to bind and enter target cells.

(142) Preferably, viral vectors comprising the improved LAMP Constructs described herein are modified from wild-type viral genomes to disable the growth of the virus in a target cell while enabling the virus to grow in a host cell (e.g., such as a packaging or helper cell) used to prepare infectious particles. Vector nucleic acids generally essential cis-acting viral sequences for replication and packaging in a helper line and expression control sequences for regulating the expression of a polynucleotide being delivered to a target cell. Other viral functions are expressed in trans in specific packaging or helper cell lines as are known in the art.

(143) Preferred improved LAMP Constructs are viral vectors derived from a virus selected from the group consisting of herpes viruses, cytomegaloviruses, foamy viruses, lentiviruses, Semliki forrest virus, AAV (adeno-associated virus), poxviruses, adenovirases and retroviruses. Such viral vectors are well known in the art.

(144) In one preferred aspect, a viral vector used is an adenoviral vector. The adenoviral genome consists of a linear double-stranded DNA molecule of approximately 36 kb carrying more than about thirty genes necessary to complete the viral replication cycle. The early genes are divided into 4 regions (E1 to E4) that are essential for viral replication with the exception of the E3 region, which is believed to modulate the anti-viral host immune response. The E1 region (E1A and E1B) encodes proteins responsible for the regulation of transcription of the viral genome. Expression of the E2 region genes (E2A and E2B) leads to the synthesis of the polypeptides needed for viral replication. The proteins encoded by the E3 region prevent cytolysis by cytotoxic T cells and tumor necrosis factor (Wold and Gooding, 1991, Virology 184: 1-8). The proteins encoded by the E4 region are involved in DNA replication, late gene expression and splicing and host cell shut off (Halbert, et al., 1985, J. Virol. 56: 250-257). The late genes generally encode structural proteins contributing to the viral capsid. In addition, the adenoviral genome carries at cis-acting 5' and 3' ITRs (Inverted Terminal Repeat) and packaging sequences essential for DNA replication. The ITRs harbor origins of DNA replication while the encapsidation region is required for the packaging of adenoviral DNA into infectious particles.

(145) Adenoviral vectors can be engineered to be conditionally replicative (CRAd vectors) in order to replicate selectively in specific cells (e.g., such as proliferative cells) as described in Heise and Kim (2000, J. Clin. Invest. 105: 847-851). In another aspect, an adenoviral vector is replication-defective for the E1 function (e.g., by total or partial deletion or mutagenesis of E1). The adenoviral backbone of the vector may comprise additional modifications (deletions, insertions or mutations in one or more viral genes). An example of an E2 modification is illustrated by the thermosensitive mutation localized on the DBP (DNA Binding Protein) encoding gene (Ensinger et

al., 1972, J. Virol. 10: 328-339). The adenoviral sequence may also be deleted of all or part of the E4 region (see, e.g., EP 974 668; Christ, et al., 2000, Human Gene Ther. 11: 415-427; Lusky, et al., 1999, J. Virol. 73: 8308-8319). Additional deletions within the non-essential E3 region may allow the size of the polynucleotide being delivered to be increased (Yeh, et al., 1997, FASEB Journal 11: 615 623). However, it may be advantageous to retain all or part of the E3 sequences coding for polypeptides (e.g., such as gp19k) allowing the virus to escape the immune system (Gooding, et al., 1990, Critical Review of Immunology 10: 53-71) or inflammatory reactions (EP 00440267.3).

(146) Second generation vectors retaining the ITRs and packaging sequences and comprising substantial genetic modifications to abolish the residual synthesis of the viral antigens also may be used in order to improve long-term expression of the expressed gene in the transduced cells (see, e.g., WO 94/28152; Lusky, et al., 1998, J. Virol 72: 2022-2032).

(147) The improved LAMP Constructs being introduced into the cell may be inserted in any location of the viral genome, with the exception of the cis-acting sequences. Preferably, it is inserted in replacement of a deleted region (E1, E3 and/or E4), preferably, within a deleted E1 region.

(148) Adenoviruses can be derived from any human or animal source, in particular canine (e.g. CAV-1 or CAV-2 Genbank ref. CAVIGENOM and CAV77082, respectively), avian (Genbank ref AAVEDSDNA), bovine (such as BAV3; Reddy, et al., 1998, J. Virol. 72: 1394 1402), murine (Genbank ref ADRMUSMAVI), ovine, feline, porcine or simian sources or alternatively, may be a hybrid virus. Any serotype can be employed. However, the human adenoviruses of the C sub-group are preferred, especially adenoviruses 2 (Ad2) and 5 (Ad5). Such viruses are available, for example, from the ATCC.

(149) Adenoviral particles or empty adenoviral capsids also can be used to transfer improved LAMP Constructs by a virus-mediated cointernalization process as described in U.S. Pat. No. 5,928,944. This process can be accomplished in the presence of cationic agent(s) such as polycarbenes or lipid vesicles comprising one or more lipid layers.

(150) Adenoviral particles may be prepared and propagated according to any conventional technique in the field of the art (e.g., WO 96/17070) using a complementation cell line or a helper virus, which supplies in trans the missing viral genes necessary for viral replication. The cell lines 293 (Graham et al., 1977, J. Gen. Virol. 36: 59-72) and PERC6 (Fallaux et al., 1998, Human Gene Therapy 9: 1909-1917) are commonly used to complement E1 deletions. Other cell lines have been engineered to complement defective vectors (Yeh, et al., 1996, J. Virol. 70: 559-565; Kroughak and Graham, 1995, Human Gene Ther. 6: 1575-1586; Wang, et al., 1995, Gene Ther. 2: 775-783; Lusky, et al., 1998, J. Virol. 72: 2022-203; EP 919627 and WO 97/04119). The adenoviral particles can be recovered from the culture supernatant but also from the cells after lysis and optionally further purified according to standard techniques (e.g., chromatography, ultracentrifugation, as described in WO 96/27677, WO 98/00524 WO 98/26048 and WO 00/50573).

(151) Cell-type specific targeting may be achieved with vectors derived from adenoviruses having a broad host range by the modification of viral surface proteins. For example, the specificity of infection of adenoviruses is determined by the attachment to cellular receptors present at the surface of permissive cells. In this regard, the fiber and penton present at the surface of the adenoviral capsid play a critical role in cellular attachment (Defer, et al., 1990, J. Virol. 64: 3661-3673). Thus, cell targeting of adenoviruses can be carried out by genetic modification of the viral gene encoding fiber and/or penton, to generate modified fiber and/or penton capable of specific interaction with unique cell surface receptors. Examples of such modifications are described in Wickam, et al., 1997, J. Virol. 71: 8221-8229; Arriberg, et al., 1997, Virol. Chem 268: 6866-6869; Roux, et al., 1989, Proc. Natl. Acad. Sci. USA 86: 9079-9083; Miller and Vile, 1995, FASEB J. 9: 190-199; WO 93/09221, and in WO 95/28494.

(152) In a particularly preferred aspect, adeno-associated viral sequences are used as vectors. Vectors derived from the human parvovirus AAV-2 (adeno-associated virus type 2) are among the

most promising gene delivery vehicles currently being developed. Several of the features of this system for packaging a single-stranded DNA suggest it as a possible alternative to naked DNA for delivery. A primary attractive feature, in contrast to other viral vectors such as vaccinia or adenovirus, is that AAV vectors do not express any viral genes. The only viral DNA sequences included in the vaccine construct are the 145 bp inverted terminal repeats (ITR). Thus, as in immunization with naked DNA, the only gene expressed is that of the allergen, or allergen chimera. Additionally, AAV vectors are known to transduce both dividing and non-dividing cells, such as human peripheral blood monocyte-derived dendritic cells, with persistent transgene expression, and with the possibility of oral and intranasal delivery for generation of mucosal immunity. Moreover, the amount of DNA required appears to be much less by several orders of magnitude, with maximum responses at doses of 10^{10} to 10^{11} particles or copies of DNA in contrast to naked DNA doses of 50 μ g or about 10^{15} copies.

(153) In one aspect, AAV vectors are packaged by co-transfection of a suitable cell line (e.g., human 293 cells) with the DNA contained in the AAV ITR chimeric protein encoding constructs and an AAV helper plasmid ACG2 containing the AAV coding region (AAV rep and cap genes) without the ITRs. The cells are subsequently infected with the adenovirus Ad5. Vectors can be purified from cell lysates using methods known in the art (e.g., such as cesium chloride density gradient ultracentrifugation) and are validated to ensure that they are free of detectable replication-competent AAV or adenovirus (e.g., by a cytopathic effect bioassay). AAV titer may be determined by quantitative PCR with virus DNA samples prepared after digestion with proteinase K. Preferably, vector titers produced by such a method are approximately 5×10^{12} to 1×10^{13} DNase resistant particles per ml.

(154) In other aspects, retroviral vectors are used. Retroviruses are a class of integrative viruses which replicate using a virus-encoded reverse transcriptase, to replicate the viral RNA genome into double stranded DNA which is integrated into chromosomal DNA of the infected cells (e.g., target cells). Such vectors include those derived from murine leukemia viruses, especially Moloney (Gilboa, et al., 1988, Adv. Exp. Med. Biol. 241: 29) or Friend's FB29 strains (WO 95/01447). Generally, a retroviral vector is deleted of all or part of the viral genes gag, pol and env and retains 5' and 3' LTRs and an encapsidation sequence. These elements may be modified to increase expression level or stability of the retroviral vector. Such modifications include the replacement of the retroviral encapsidation sequence by one of a retrotransposon such as VL30 (see, e.g., U.S. Pat. No. 5,747,323). Preferably, the improved LAMP Construct is inserted downstream of the encapsidation sequence, preferably in opposite direction relative to the retroviral genome. Cell specific targeting may be achieved by the conjugation of antibodies or antibody fragments to the retroviral envelope protein as is known in the art.

(155) Retroviral particles are prepared in the presence of a helper virus or in an appropriate complementation (packaging) cell line which contains integrated into its genome the retroviral genes for which the retroviral vector is defective (e.g. gag/pol and env). Such cell lines are described in the prior art (Miller and Rosman, 1989, BioTechniques 7: 980; Danos and Mulligan, 1988, Proc. Natl. Acad. Sci. USA 85: 6460; Markowitz, et al., 1988, Virol. 167: 400). The product of the env gene is responsible for the binding of the viral particle to the viral receptors present on the surface of the target cell and, therefore determines the host range of the retroviral particle. In the context of the invention, it is advantageous to use a packaging cell line, such as the PA317 cells (ATCC CRL 9078) or 293E16 (WO97/35996) containing an amphotropic envelope protein, to allow infection of human and other species' target cells. The retroviral particles are preferably recovered from the culture supernatant and may optionally be further purified according to standard techniques (e.g. chromatography, ultracentrifugation).

(156) Other suitable viruses include poxviruses. The genome of several members of poxyiridae has been mapped and sequenced. A poxyviral vector may be obtained from any member of the poxyiridae, in particular canarypox, fowlpox and vaccinia virus. Suitable vaccinia viruses include,

but are not limited to, the Copenhagen strain (Goebel, et al., 1990, Virol. 179: 247-266; Johnson, et al., 1993, Virol. 196: 381-401), the Wyeth strain and the modified Ankara (MVA) strain (Antoine, et al., 1998, Virol. 244: 365-396). The general conditions for constructing a vaccinia virus vector are known in the art (see, e.g., EP 83 286 and EP 206 920; Mayr et al., 1975, Infection 3: 6-14; Sutter and Moss, 1992, Proc. Natl. Acad. Sci. USA 89: 10847-10851). Preferably, the polynucleotide of interest is inserted within a non-essential locus such as the noncoding intergenic regions or any gene for which inactivation or deletion does not significantly impair viral growth and replication.

(157) Poxviral particles are prepared as described in the art (Piccini, et al., 1987, Methods of Enzymology 153: 545-563; U.S. Pat. Nos. 4,769,330; 4,772,848; 4,603,112; 5,100,587 and 5,179,993). Generally, a donor plasmid is constructed, amplified by growth in *E. coli* and isolated by conventional procedures. Then, it is introduced into a suitable cell culture (e.g. chicken embryo fibroblasts) together with a poxvirus genome, to produce, by homologous recombination, poxviral particles. These can be recovered from the culture supernatant or from the cultured cells after a lysis step (e.g., chemical lysis, freezing/thawing, osmotic shock, sonication and the like). Consecutive rounds of plaque purification can be used to remove contaminating wild type virus. Viral particles can then be purified using the techniques known in the art (e.g., chromatographic methods or ultracentrifugation on cesium chloride or sucrose gradients).

(158) The use of vaccinia as a live virus vaccine in the global campaign to eradicate smallpox made vaccinia an obvious choice for development as a live recombinant vaccine vector. Live recombinant vaccinia viruses expressing close to 100 different foreign proteins have been reported, and a number of these are effective experimental vaccines (reviewed by Moss and Flexner, 1987). Vaccinia is particularly versatile as an expression vector because of its large genomic size, capability of accepting at least 25,000 base pairs of foreign DNA, and its ability to infect most eukaryotic cell types, including insect cells (ibid.). Unlike other DNA viruses, poxviruses replicate exclusively in the cytoplasm of infected cells, reducing the possibility of genetic exchange of recombinant viral DNA with the host chromosome. Recombinant vaccinia vectors have been shown to properly process and express proteins from a variety of sources including man, other mammals, parasites, RNA and DNA viruses, bacteria and bacteriophage.

(159) The expression of DNA encoding a foreign protein is controlled by host virus regulatory elements, including upstream promoter sequences and, where necessary, RNA processing signals. Insertion of foreign DNA into nonessential regions of the vaccinia virus genome has been carried out by homologous recombination (Panicali, et al., Proc. Nat'l. Acad. Sci. USA, 79: 4927, 1982; Mackett, et al., Proc. Nat'l. Acad. Sci. USA, 79: 7415, 1982).

(160) Expression of allergens by the improved LAMP Construct may occur because of transcriptional regulatory elements at or near the site of insertion or by more precise genetic engineering. Plasmid vectors that greatly facilitate insertion and expression of foreign genes have been constructed (Mackett, et al., J. Virol, 49: 857, 1982). These vectors contain an expression site, composed of a vaccinia transcriptional promoter and one or more unique restriction endonuclease sites for insertion of the foreign coding sequence flanked by DNA from a nonessential region of the vaccinia genome. The choice of promoter determines both the time (e.g., early or late) and level of expression, whereas the flanking DNA sequence determines the site of homologous recombination.

(161) Only about one in a thousand virus particles produced by this procedure is a recombinant. Although recombinant virus plaques can be identified by DNA hybridization, efficient selection procedures have been developed. By using segments of nonessential vaccinia virus thymidine kinase (TK) gene as flanking sequences, the foreign gene recombines into the TK locus and by insertion inactivates the TK gene. Selection of TK virus is achieved by carrying out the virus plaque assay in TK cells in the presence of 5-bromodeoxyuridine. Phosphorylation of the nucleoside analogue and consequent lethal incorporation into viral DNA occurs only in cells infected with TK+parental virus. Depending on the efficiency of the transfection and

recombination, up to 80 of the plaques are desired recombinants, and the rest are spontaneous TK mutants. Plasmid vectors that contain the *E. coli* beta-galactosidase gene, as well as an expression site for a second gene, permit an alternative method of distinguishing recombinant from parental virus (Chakrabarti, et al., Mol. Cell. Biol., 5: 3403, 1985). Plaques formed by such recombinants can be positively identified by the blue color that forms upon addition of an appropriate indicator. By combining both TK selection and beta-galactosidase expression, recombinant virus is readily and quickly isolated. The recombinants are then amplified by propagation in suitable cell lines and expression of the inserted gene is checked by appropriate enzymological, immunological or physical procedures.

(162) An upper limit to the amount of genetic information that can be added to the vaccinia virus genome is not yet known. However, the addition of nearly 25,000 base pairs of foreign DNA had no apparent deleterious effect on virus yield (Smith, et al., Gene, 25:21, 1983). Were it necessary, large segments of the vaccinia virus genome could be deleted to provide additional capacity (Moss, et al., J. Virol. 40: 387, 1981).

(163) Viral capsid molecules may include targeting moieties to facilitate targeting and/or entry into cells. Suitable targeting molecules, include, but are not limited to: chemical conjugates, lipids, glycolipids, hormones, sugars, polymers (e.g. PEG, polylysine, PEI and the like), peptides, polypeptides (see, e.g., WO 94/40958), vitamins, antigens, lectins, antibodies and fragments thereof. Preferably, such targeting molecules recognize and bind to cell-specific markers, tissue-specific markers, cellular receptors, viral antigens, antigenic epitopes or tumor-associated markers.

(164) Compositions comprising an improved LAMP Construct based on viral particles may be formulated in the form of doses of between 10 and 10^{sup.14} i.u. (infectious units), and preferably, between 10 and 10^{sup.11} i.u. The titer may be determined by conventional techniques. The doses of LAMP Constructs are preferably comprised between 0.01 and 10 mg/kg, more especially between 0.1 and 2 mg/kg.

(165) Self-Replicating RNA

(166) Self-replicating RNA virus vectors can also be constructed using the improved LAMP Constructs as described herein. For example, alphaviruses, flaviviruses, measles virus and rhabdoviruses can be used to generate self-replicating RNA virus vaccines. Preferred strains of self-replicating RNA viruses include, but are not limited to rabies virus (RABV), vesicular stomatitis virus (VSV), West Nile virus, Kunjin virus, Semliki Forest virus (SFV), Sindbis virus (SIN) and/or Venezuelan equine encephalitis virus (VEE).

(167) Self-replicating RNA viruses express the native antigen upon delivery into tissue, thus mimicking live attenuated vaccines without having the risk of reversion to pathogenicity. They also stimulate the innate immune system, thus potentiating responses. See, e.g., Ljungberg, K. “*Self-replicating alphavirus RNA vaccines*,” Expert Rev Vaccines (2):177-94 (2015); Lundstrom, K., “*Oncolytic Alphaviruses in Cancer Immunotherapy*”, Vaccines 5:9 (2017); Lundstrom, K. “*Replicon RNA Viral Vectors as Vaccines*,” Vaccines 4:39 (2016) (hereby incorporated by reference in their entirety). Use of self-replicating vaccines comprising the improved LAMP Constructs described herein can also be used in prime-boost protocols.

(168) Moreover, self-replicating RNA viruses can also be encapsulated by liposomes, as described herein, to improve delivery and targeting. Immunization with self-replicating RNA viruses comprising the improved LAMP Constructs described herein may provide higher transient expression levels of allergens resulting in generation of neutralizing antibody responses and protection against lethal challenges under safe conditions.

(169) Cell-Based Delivery Vehicles

(170) The improved LAMP Constructs according to the invention can be delivered to target cells by means of other cells (“delivery cells”) which comprise the constructs. Methods for introducing constructs into cells are known in the art and include microinjection of DNA into the nucleus of a cell (Capechi, et al., 1980, Cell 22: 479-488); transfection with CaPO₄ (Chen and Okayama,

1987, Mol. Cell Biol. 7: 2745-2752), electroporation (Chu, et al., 1987, Nucleic Acid Res. 15: 1311-1326); lipofection/liposome fusion (Feigner, et al., 1987, Proc. Natl. Acad. Sci. USA 84: 7413-7417) and particle bombardment (Yang, et al., 1990, Proc. Natl. Acad. Sci. USA 87: 9568-9572). Suitable cells include autologous and non-autologous cells, and may include xenogenic cells. Delivery cells may be induced to deliver their contents to the target cells by inducing their death (e.g., by providing inducible suicide genes to these cells).

(171) Accessory Molecules

(172) The compositions comprising the improved LAMP Constructs according to the invention may comprise one or more accessory molecules for facilitating the introduction of an improved LAMP Construct into a cell and/or for enhancing a particular therapeutic effect and/or enhancing antibody production.

(173) In addition, the composition comprising the improved LAMP Construct according to the present invention may include one or more stabilizing substance(s), such as lipids, nuclease inhibitors, hydrogels, hyaluronidase (WO 98/53853), collagenase, polymers, chelating agents (EP 890362), in order to inhibit degradation within the animal/human body and/or improve transfection/infection of the vector into a target cell. Such substances may be used alone or in combination (e.g., cationic and neutral lipids).

(174) It has also been shown that adenovirus proteins are capable of destabilizing endosomes and enhancing the uptake of DNA into cells. The mixture of adenoviruses to solutions containing a lipid-complexed DNA vector or the binding of DNA to polylysine covalently attached to adenoviruses using protein cross-linking agents may substantially improve the uptake and expression of an improved LAMP Construct (see, e.g., Curiel, et al., 1992, Am. J. Respir. Cell. Mol. Biol. 6: 247-252).

(175) Host Cells

(176) Improved LAMP Constructs according to the invention can be expressed in a variety of host cells, including, but not limited to: prokaryotic cells (e.g., *E. coli*, *Staphylococcus* sp., *Bacillus* sp.); yeast cells (e.g., *Saccharomyces* sp.); insect cells; nematode cells; plant cells; amphibian cells (e.g., *Xenopus*); avian cells; and mammalian cells (e.g., human cells, mouse cells, mammalian cell lines, primary cultured mammalian cells, such as from dissected tissues).

(177) The molecules can be expressed in host cells isolated from an organism, host cells which are part of an organism, or host cells which are introduced into an organism. In one aspect, improved LAMP Constructs are expressed in host cells in vitro, e.g., in culture. In another aspect, improved LAMP Constructs are expressed in a transgenic organism (e.g., a transgenic mouse, rat, rabbit, pig, primate, etc.) that comprises somatic and/or germline cells comprising nucleic acids encoding the improved LAMP Constructs. Methods for constructing transgenic animals are well known in the art and are routine.

(178) Improved LAMP Constructs also can be introduced into cells in vitro, and the cells (e.g., such as stem cells, hematopoietic cells, lymphocytes, and the like) can be introduced into the host organism. The cells may be heterologous or autologous with respect to the host organism. For example, cells can be obtained from the host organism, improved LAMP Constructs introduced into the cells in vitro, and then reintroduced into the host organism.

(179) Antigen Presenting Cells

(180) In a preferred aspect of the invention, an improved LAMP Construct as described herein is introduced into a natural or engineered antigen presenting cell.

(181) The term "antigen presenting cell" (APC) as used herein intends any cell which presents on its surface an antigen (e.g., an allergen) in association with a major histocompatibility complex molecule, preferably a class II molecule, or portion thereof. Examples of suitable APCs are discussed in detail below and include, but are not limited to, whole cells such as macrophages, dendritic cells, B cells, hybrid APCs, and foster antigen presenting cells. Methods of making hybrid APCs are described and known in the art.

(182) Dendritic cells (DCs) are potent antigen-presenting cells. It has been shown that DCs provide all the signals required for T cell activation and proliferation. These signals can be categorized into two types. The first type, which gives specificity to the immune response, is mediated through interaction between the T-cell receptor/CD3 ("TCR/CD3") complex and an allergenic peptide presented by a major histocompatibility complex ("MHC" defined above) class I or II protein on the surface of APCs. This interaction is necessary, but not sufficient, for T cell activation to occur. In fact, without the second type of signals, the first type of signals can result in T cell anergy. The second type of signals, called co-stimulatory signals, is neither antigen-specific nor MHC-restricted, and can lead to a full proliferation response of T cells and induction of T cell effector functions in the presence of the first type of signals.

(183) Several molecules have been shown to enhance co-stimulatory activity. These include, but are not limited to, heat stable antigen (HSA), chondroitin sulfate-modified MHC invariant chain (Ii-CS), intracellular adhesion molecule I (ICAM-1), and B7 co-stimulatory molecule on the surface of APCs and its counter-receptor CD28 or CTLA-4 on T cells.

(184) Other important co-stimulatory molecules are CD40, CD54, CD80, CD86. As used herein, the term "co-stimulatory molecule" encompasses any single molecule or combination of molecules which, when acting together with a peptide/MHC complex bound by a TCR on the surface of a T cell, provides a co-stimulatory effect which achieves activation of the T cell that binds the peptide. The term thus encompasses B7, or other co-stimulatory molecule(s) on an APC, fragments thereof (alone, complexed with another molecule(s), or as part of a fusion protein) which, together with peptide/MHC complex, binds to a cognate ligand and result in activation of the T cell when the TCR on the surface of the T cell specifically binds the peptide. Co-stimulatory molecules are commercially available from a variety of sources, including, for example, Beckman Coulter.

(185) In one aspect of the invention, the method described in Romani et al., *J. Immunol. Methods* 196: 135-151, 1996, and Bender et al, *J. Immunol. Methods* 196: 121-135, 1996, are used to generate both immature and mature dendritic cells from the peripheral blood mononuclear cells (PBMCs) of a mammal, such as a murine, simian or human. Briefly, isolated PBMCs are pre-treated to deplete T- and B-cells by means of an immunomagnetic technique. Lymphocyte-depleted PBMC are then cultured for in RPMI medium 9 e.g., about 7 days), supplemented with human plasma (preferably autologous plasma) and GM-CSF/IL-4, to generate dendritic cells. Dendritic cells are nonadherent when compared to their monocyte progenitors. Thus, on approximately day 7, non-adherent cells are harvested for further processing.

(186) The dendritic cells derived from PBMC in the presence of GM-CSF and IL-4 are immature, in that they can lost the nonadherence property and revert back to macrophage cell fate if the cytokine stimuli are removed from the culture. The dendritic cells in an immature state are very effective in processing native protein antigens for the MHC class II restricted pathway (Romani, et al., *J. Exp. Med.* 169:1169, 1989). Further maturation of cultured dendritic cells is accomplished by culturing for 3 days in a macrophage-conditioned medium (CM), which contains the necessary maturation factors. Mature dendritic cells are less able to capture new proteins for presentation but are much better at stimulating resting T cells (both CD4 and CD8) to grow and differentiate.

(187) Mature dendritic cells can be identified by their change in morphology, such as the formation of more motile cytoplasmic processes; by their nonadherence; by the presence of at least one of the following markers: CD83, CD68, HLA-DR or CD86; or by the loss of Fc receptors such as CD 115 (reviewed in Steinman, *Annu. Rev. Immunol.* 9: 271, 1991). Mature dendritic cells can be collected and analyzed using typical cytofluorography and cell sorting techniques and devices, such as FACScan and FACStar. Primary antibodies used for flow cytometry are those specific to cell surface antigens of mature dendritic cells and are commercially available. Secondary antibodies can be biotinylated Igs followed by FITC- or PE-conjugated streptavidin.

(188) Alternatively, others have reported that a method for upregulating (activating) dendritic cells and converting monocytes to an activated dendritic cell phenotype. This method involves the

addition of calcium ionophore to the culture media convert monocytes into activated dendritic cells. Adding the calcium 21 ionophore A23187, for example, at the beginning of a 24-48 hour culture period resulted in uniform activation and dendritic cell phenotypic conversion of the pooled "monocyte plus DC" fractions: characteristically, the activated population becomes uniformly CD 14 (Leu M3) negative, and upregulates HLA-DR, HLA-DQ, ICAM-1, 137.1, and 137.2. Furthermore, this activated bulk population functions as well on a small numbers basis as a further purified. Specific combination(s) of cytokines have been used successfully to amplify (or partially substitute) for the activation/conversion achieved with calcium ionophore: these cytokines include but are not limited to G-CSF, GM-CSF, IL-2, and IL-4. Each cytokine when given alone is inadequate for optimal upregulation.

(189) The second approach for isolating APCs is to collect the relatively large numbers of precommitted APCs already circulating in the blood. Previous techniques for isolating committed APCs from human peripheral blood have involved combinations of physical procedures such as metrizamide gradients and adherence/nonadherence steps (Freudenthal et al. PNAS 87: 7698-7702, 1990); Percoll gradient separations (Mehta-Damani, et al., J. Immunol. 153: 996-1003, 1994); and fluorescence activated cell sorting techniques (Thomas et al., J. Immunol. 151: 6840-52, 1993).

(190) There are many other methods routine in the art for isolating professional antigen presenting cells (or their precursors) and that such methods and others which may be developed are not limiting and are encompassed within the scope of the invention.

(191) In one embodiment, the APCs and therefore the cells presenting one or more allergens as described herein are autologous. In another embodiment, the APCs presenting the allergens as described herein are allogeneic, i.e., derived from a different subject.

(192) As discussed herein, improved LAMP Constructs can be introduced into APCs using the methods described above or others known in the art, including, but not limited to, transfection, electroporation, fusion, microinjection, viral-based delivery, or cell based delivery. Arthur et al., Cancer Gene Therapy 4(1): 17-25, 1997, reports a comparison of gene transfer methods in human dendritic cells.

(193) Known, partial and putative human leukocyte antigen (HLA), the genetic designation for the human MHC, amino acid and nucleotide sequences, including the consensus sequence, are published (see, e.g., Zemmour and Parham, Immunogenetics 33: 310-320, 1991), and cell lines expressing HLA variants are known and generally available as well, many from the American Type Culture Collection ("ATCC"). Therefore, using PCR, MHC class II-encoding nucleotide sequences are readily operatively linked to an expression vector of this invention that is then used to transform an appropriate cell for expression therein.

(194) Professional APCs can be used, such as macrophages, B cells, monocytes, dendritic cells, and Langerhans cells. These are collected from the blood or tissue of 1) an autologous donor; 2) a heterologous donor having a different HLA specificity than the host to be treated; or 3) from a xenogeneic donor of a different species using standard procedures (Coligan, et. al., Current Protocols in Immunology, sections 3 and 14, 1994). The cells may be isolated from a normal host or a patient having an infectious disease, cancer, autoimmune disease, or allergy.

(195) Professional APCs may be obtained from the peripheral blood using leukopheresis and "FICOLL/HYPAQUE" density gradient centrifugation (stepwise centrifugation through Ficoll and discontinuous Percoll density gradients). Procedures are utilized which avoid the exposure of the APCs to allergens which could be internalized by the APCs, leading to activation of T cells not specific for the Allergen X (SEQ ID NO:Y) of interest.

(196) Cells which are not naturally antigen presenting can be engineered to be antigen presenting by introducing sequences encoding appropriate molecules. For example, nucleic acid sequences encoding MHC class II molecules, accessory molecules, co-stimulatory molecules and antigen processing assisting molecules can be introduced after direct synthesis, cloning, purification of DNA from cells containing such genes, and the like. One expedient means to obtain genes for

encoding the molecules used in the improved LAMP Constructs and methods described herein is by polymerase chain reaction (PCR) amplification on selected nucleic acid templates with selected oligonucleotide primer pairs. For example, epithelial cells, endothelial cells, tumor cells, fibroblasts, activated T cells, eosinophils, keratinocytes, astrocytes, microglial cells, thymic cortical epithelial cells, Schwann cells, retinal pigment epithelial cells, myoblasts, vascular smooth muscle cells, chondrocytes, enterocytes, thyrocytes and kidney tubule cells can be used. These may be primary cells recently explanted from a host and not extensively passaged in cell culture to form an established cell line, or established cell lines that are relatively homogeneous and capable of proliferating for many generations or indefinitely.

(197) Cells that are not professional APCs are isolated from any tissue of an autologous donor; a heterologous donor or a xenogeneic donor, where they reside using a variety of known separation methods (Darling, *Animal Cells: Culture and Media*. J. Wiley, New York, 1994; Freshney, *Culture of Animal Cells*. Alan R. Liss, Inc., New York, 1987). Non-autologous cells, e.g., heterologous or xenogeneic cells, can be engineered ex vivo to express HLA class I and class II molecules that match known human HLA specificities. These cells can then be introduced into a human subject matching the HLA specificity of the engineered cells. The cells are further engineered ex vivo to express one or more LAMP Constructs according to the invention.

(198) The engineered cells are maintained in cell culture by standard cell culture methods (Darling, *Animal Cells: Culture and Media*". J. Wiley, New York, 1994; Freshney, *Culture of Animal Cells*". Alan R. Liss, Inc., New York, 1987). Cell lines for use in the present invention are obtained from a variety of sources (e.g., ATCC Catalogue of Cell Lines & Hybridomas, American Type Culture Collection, 8th edition, 1995), or are produced using standard methods (Freshney, *Culture of Immortalized Cells*, Wiley-Liss, New York, 1996). Non-transformed cell lines are preferred for use in human subjects.

(199) In one aspect, CD34+precursors that are differentiating under the influence of GM-CSF into dendritic cells are obtained from the body of a subject and nucleic acids encoding LAMP Constructs according to the invention are introduced into the cells, which are then injected into the subject. Utilizing the improved LAMP Constructs as described herein will enhance the association of peptides derived from a particular antigen with MHC class II molecules on the transduced antigen presenting cells, resulting in significantly more potent systemic T cell dependent immune responses and/or antibody production. While the antigen presenting cells transfected in this strategy are preferably autologous cells, any MHC class II cells that effectively present antigen in the host may be used as described above.

(200) Peptide Vaccines

(201) Also within the scope of this invention are peptide vaccines encoded by the improved LAMP Construct. Preferably, the allergen is processed within the compartment/organelle (or subsequent compartment/organelle to which it is delivered) to generate an epitope bound to an MHC class II molecule capable of modulating an immune response.

(202) The peptide vaccines encoded by the improved LAMP Constructs may also be bound in a membranous structure to facilitate its administration to the body of an organism. For example, the peptide vaccine encoded by the improved LAMP Construct may be incorporated into liposomes, as described in U.S. Pat. No. 4,448,765.

(203) When a protein or polypeptide is to be used as an immunogen, it may be produced by expression of any one or more of the improved LAMP Constructs described herein in a recombinant cell or it may be prepared by chemical synthesis. For example, the Merrifield technique (Journal of American Chemical Society, vol. 85, pp. 2149-2154, 1968), can be used.

(204) Treatment of Allergies

(205) This invention provides a formulation useful for the treatment of pollinosis correlated with Allergen X. It has previously been determined that delivering a DNA plasmid encoding the protein coding sequence of an allergen to an animal can increase IFN-gamma production and lower IL-4

production, which is useful in treating animals allergic to the specific allergen. The present invention provides improved LAMP Constructs for treating patients with an allergy correlated to Allergen X. The improved LAMP Constructs has a specific intracellular trafficking pattern that intersects with MHC class II vesicles, and results in enhanced presentation of Allergen X to the immune system, specifically resulting in an enhanced antibody response. Nucleic acids and compositions provided by the present invention are useful for conducting allergy immunotherapy. (206) The present invention provides a formulation that when administered to a cell results in an increased specific antibody response. The increased antibody response to Allergen X is useful for treating an IgE-mediated allergic disease. IgE has certain properties related to its cellular restriction and the resulting intracellular signaling upon binding cognate allergen. IgE is generated against an allergen when B cells receive IL-4 secreted by Th2 cells. This helps instruct B cells to produce IgE class antibodies. Upon secretion by B cells, IgE binds to Fc-εRI, its high affinity receptor expressed by mast cells and eosinophils, resulting in these cells and the animal becoming sensitized to future allergen exposure. Consequently, the symptoms of allergy can be triggered upon the ingestion, inhalation, or mucosal contact with an allergen. Due to the binding properties of antibodies, it has been proposed that one way of reducing allergy symptoms is to chelate free allergen available for binding by IgE through competition with other antibody classes. In particular, an allergy formulation that increases IgG has been proposed to be a pathway for reducing allergic disease. The invention described herein induces enhanced IgG production, thus causing a decrease in the ratio of IgE to IgG in a clinically significant manner.

(207) In a particularly preferred embodiment, the invention provides a method of treatment or prevention of an allergy to Allergen X by using the improved LAMP Constructs as described herein. In one preferred method, improved LAMP Constructs comprising polynucleotides encoding Allergen X described herein targets the Allergen X allergen to an endosomal/lysosomal compartment or lysosome-related organelle and for association with an MHC class II molecule either within the compartment/organelle or within another compartment/organelle to which the antigen is delivered. Such chimeric DNA molecules can encode additional domain sequences as described above (e.g., sequences encoding transmembrane domains, signal sequences, cytoplasmic domains for targeting to an endosomal/lysosomal compartment or lysosome-related organelles, dileucine domains, Tyr motif domains, proline rich domains, Ser-Val-Val domains, and the like).

(208) Host Cells

(209) In bacterial systems, a number of expression vectors may be advantageously selected depending upon the intended use. For example, when a large quantity of a protein is to be produced (e.g., to express the encoded polypeptides of the improved LAMP Construct), vectors which direct the expression of high levels of fusion protein products that are readily purified may be desirable. Such vectors include, but are not limited to, the *E. coli* expression vector pUR278 (Ruther et al., EMBO 1. 2:1791 (1983)), in which the coding sequence may be ligated individually into the vector in frame with the lac Z coding region so that a fusion protein is produced; pIN vectors (Inouye & Inouye, Nucleic Acids Res. 13:3101-3109 (1985); Van Heeke & Schuster, J. Biol. Chem. 24:5503-5509 (1989)); and the like. pGEX vectors may also be used to express foreign polypeptides as fusion proteins with glutathione S-transferase (GST). In general, such fusion proteins are soluble and can easily be purified from lysed cells by adsorption and binding to matrix glutathione agarose beads followed by elution in the presence of free glutathione. The pGEX vectors are designed to include thrombin or Factor Xa protease cleavage sites so that the cloned target gene product can be released from the GST moiety.

(210) In an insect system, *Autographa californica* nuclear polyhedrosis virus (AcNPV) may be used as a vector to express the encoded polypeptides of the improved LAMP Construct. The virus grows in *Spodoptera frugiperda* cells. Coding sequences may be cloned individually into non-essential regions (for example, the polyhedrin gene) of the virus and placed under control of an AcNPV promoter (for example, the polyhedrin promoter).

(211) In mammalian host cells, a number of viral-based expression systems may be utilized to express the encoded polypeptides of the improved LAMP Construct. In cases where an adenovirus is used as an expression vector, the coding sequence of interest may be ligated to an adenovirus transcription/translation control complex, e.g., the late promoter and tripartite leader sequence. This chimeric gene may then be inserted in the adenovirus genome by *in vitro* or *in vivo* recombination. (212) Insertion in a non-essential region of the viral genome (e.g., region E1 or E3) will result in a recombinant virus that is viable and capable of expressing the encoded polypeptides of the improved LAMP Construct in infected hosts (e.g., see Logan & Shenk, *Proc. Natl. Acad. Sci. USA* 81:355-359 (1984)).

(213) Specific initiation signals may also be required for efficient translation of inserted coding sequences. These signals include the ATG initiation codon and adjacent sequences. Furthermore, the initiation codon must be in phase with the reading frame of the desired coding sequence to ensure translation of the entire insert. These exogenous translational control signals and initiation codons can be of a variety of origins, both natural and synthetic. The efficiency of expression may be enhanced by the inclusion of appropriate transcription enhancer elements, transcription terminators, etc. (see, e.g., Bittner et al., *Methods in Enzymol.* 153:51-544 (1987)).

(214) In addition, a host cell strain may be chosen which modulates the expression of the inserted sequences, or modifies and processes the gene product in the specific fashion desired. Such modifications (e.g., glycosylation) and processing (e.g., cleavage) of protein products may be important for the function of the protein. Different host cells have characteristic and specific mechanisms for the post-translational processing and modification of proteins and gene products. Appropriate cell lines or host systems can be chosen to ensure the correct modification and processing of the foreign protein expressed, to this end, eukaryotic host cells which possess the cellular machinery for proper processing of the primary transcript, glycosylation, and phosphorylation of the gene product may be used. Such mammalian host cells include, but are not limited to, CHO, VERO, BHK, HeLa, COS, NSO, MDCK, 293, 3T3, and W138.

(215) For long-term, high-yield production of recombinant proteins, stable expression is preferred. For example, cell lines which stably express the encoded polypeptides of the improved LAMP Construct may be engineered. Rather than using expression vectors which contain viral origins of replication, host cells can be transformed with a polynucleotide controlled by appropriate expression control elements (e.g., promoter, enhancer, sequences, transcription terminators, polyadenylation sites, etc.), and a selectable marker. Following the introduction of the foreign polynucleotide, engineered cells may be allowed to grow for 1-2 days in an enriched media, and then are switched to a selective media. The selectable marker in the recombinant plasmid confers resistance to the selection and allows cells to stably integrate the plasmid into their chromosomes and grow to form foci which in turn can be cloned and expanded into cell lines. This method may advantageously be used to engineer cell lines which express the encoded polypeptides of the improved LAMP Construct.

(216) A number of selection systems may be used, including but not limited to, the herpes simplex virus thymidine kinase (Wigler et al., *Cell* 11:223 (1977)), hypoxanthine-guanine phosphoribosyltransferase (Szybalska & Szybalski, *Proc. Natl. Acad. Sci. USA* 48:202 (1992)), and adenine phosphoribosyltransferase (Lowy et al., *Cell* 22:817 (1980)) genes can be employed in tk-, hgp^rt- or ap^rt-cells, respectively. Also, antimetabolite resistance can be used as the basis of selection for the following genes: dhfr, which confers resistance to methotrexate (Wigler et al., *Natl. Acad. Sci. USA* 77:357 (1980); O'Hare et al., *Proc. Natl. Acad. Sci. USA* 78:1527 (1981)); gpt, which confers resistance to mycophenolic acid (Mulligan & Berg, *Proc. Natl. Acad. Sci. USA* 78:2072 (1981)); neo, which confers resistance to the aminoglycoside G-418 (Goldspiel et al., *Clinical Pharmacy*, 12: 488-505 (1993); Wu and Wu, *Biotherapy* 3:87-95 (1991); Tolstoshev, *Ann. Rev. Pharmacol. Toxicol.* 32:573-596 (1993); Mulligan, *Science* 260:926-932 (1993); and Morgan and Anderson, *Ann. Rev. Biochem.* 62: 191-217 (1993); TIB TECH 11(5):155-215 (May; 1993));

and hygromycin, which confers resistance to hygromycin (Santerre et al., Gene 30:147 (1984)). Methods commonly known in the art of recombinant DNA technology may be routinely applied to select the desired recombinant clone, and such methods are described, for example; in Ausubel et al. (eds.), Current Protocols in Molecular Biology, John Wiley & Sons, NY (1993); Kriegler, Gene Transfer and Expression, A Laboratory Manual, Stockton Press, NY (1990); and in Chapters 12 and 13, Dracopoli et al. (eds), Current Protocols in Human Genetics, John Wiley & Sons, NY (1994); Colberre-Garapin et al., J. Mol. Biol. 150:1 (1981).

(217) The expression levels of the encoded polypeptides of the improved LAMP Construct can be increased by vector amplification (for a review, see Bebbington and Hentschel, *The Use Of Vectors Based On Gene Amplification For The Expression Of Cloned Genes In Mammalian Cells In DNA Cloning*, Vol. 3. (Academic Press, New York, 1987)). When a marker in the vector system expressing the encoded polypeptides of the improved LAMP Construct is amplifiable, an increase in the level of inhibitor present in the host cell culture will increase the number of copies of the marker gene. Since the amplified region is associated with the coding sequence, production of the encoded polypeptides of the improved LAMP Construct will also increase (Crouse et al., Mol. Cell. Biol. 3:257 (1983)).

(218) Other elements that can be included in vector sequences include heterologous signal peptides (secretion signals), membrane anchoring sequences, introns, alternative splice sites, translation start and stop signals, inteins, biotinylation sites and other sites promoting post-translational modifications, purification tags, sequences encoding fusions to other proteins or peptides, separate coding regions separated by internal ribosome reentry sites, sequences encoding "marker" proteins that, for example, confer selectability (e.g., antibiotic resistance) or sortability (e.g., fluorescence), modified nucleotides, and other known polynucleotide cis-acting features not limited to these examples.

(219) Once the encoded polypeptides of the improved LAMP Construct has been produced by recombinant expression, it may be purified by any method known in the art for purification of a protein, for example, by chromatography (e.g., ion exchange, affinity (particularly by Protein A affinity and immunoaffinity for the specific allergen), and sizing column chromatography), centrifugation, differential solubility, or by any other standard technique for the purification of proteins. Further, the encoded polypeptides of the improved LAMP Construct may be fused to heterologous polypeptide sequences described herein or otherwise known in the art to facilitate purification.

(220) Administration

(221) Vaccine material according to this invention may contain the immune stimulatory improved LAMP Constructs described herein or may be recombinant microorganisms, or antigen presenting cells which express the immune stimulatory improved LAMP Constructs. Preparation of improved LAMP Constructs containing vaccine material according to this invention and administration of such improved LAMP Constructs for immunization of individuals are accomplished according to principles of immunization that are well known to those skilled in the art.

(222) Large quantities of these materials may be obtained by culturing recombinant or transformed cells containing replicons that express the improved LAMP Constructs described herein. Culturing methods are well-known to those skilled in the art and are taught in one or more of the documents cited above. The improved LAMP Construct vaccines are generally produced by culture of recombinant or transformed cells and formulated in a pharmacologically acceptable solution or suspension, which is usually a physiologically-compatible aqueous solution, or in coated tablets, tablets, capsules, suppositories or ampules, as described in the art, for example in U.S. Pat. No. 4,446,128, incorporated herein by reference. Administration may be any suitable route, including oral, rectal, intranasal or by injection where injection may be, for example, transdermal, subcutaneous, intramuscular or intravenous.

(223) The improved LAMP Constructs are administered to a mammal in an amount sufficient to

induce an immune response in the mammal. A minimum preferred amount for administration is the amount required to elicit antibody formation to a concentration at least 4 times that which existed prior to administration. A typical initial dose for administration would be 10-5000 micrograms when administered intravenously, intramuscularly or subcutaneously, or 10.sup.5 to 10.sup.11 plaque forming units of a recombinant vector, although this amount may be adjusted by a clinician doing the administration as commonly occurs in the administration of vaccines and other agents which induce immune responses. A single administration may usually be sufficient to induce immunity, but multiple administrations may be carried out to assure or boost the response.

(224) The improved LAMP Construct vaccines maybe tested initially in a non-human mammal (e.g., a mouse or primate). For example, assays of the immune responses of inoculated mice can be used to demonstrate greater antibody, T cell proliferation, and cytotoxic T cell responses to the improved LAMP Constructs than to wild type allergen. Improved LAMP Constructs can be evaluated in Rhesus monkeys to determine whether the vaccine formulation that is highly effective in mice will also elicit an appropriate monkey immune response. In one aspect, each monkey receives a total of 5 mg DNA per immunization, delivered IM and divided between 2 sites, with immunizations at day 0 and at weeks 4, 8, and 20, with additional doses optional. Antibody responses, ADCC, CD4+ and CD8+ T-cell cytokine production, CD4+ and CD8+ T-cell antigen-specific cytokine staining can be measured to monitor immune responses to the vaccine.

(225) Further description of suitable methods of formulation and administration according to this invention may be found in U.S. Pat. No. 4,454,116 (constructs), U.S. Pat. No. 4,681,762 (recombinant bacteria), and U.S. Pat. Nos. 4,592,002 and 4,920,209 (recombinant viruses).

(226) Kits

(227) The invention further comprises kits to facilitate performing the methods described herein. In one aspect, a kit comprises an improved LAMP Construct as described herein and a cell for receiving the improved LAMP Construct. The kit may additionally comprise one or more nucleic acids for engineering the cell into a professional APC. In one aspect, however, the cell is a professional APC. The cell may or may not express co-stimulatory molecules. In a preferred aspect, when the cell does not express co-stimulatory molecules, the allergen encoded by the improved LAMP Construct is an autoantigen. In another aspect, a panel of cells is provided expressing different MHC molecules (e.g., known to be expressed in human beings). In a further aspect, the kit comprises reagents to facilitate entry of the improved LAMP Constructs into a cell (e.g., lipid-based formulations, viral packaging materials, cells, and the like). In still a further aspect, one or more T cell lines specific for the allergen encoded by the improved LAMP Construct is provided, to verify the ability of the improved LAMP Construct to elicit, modulate, or enhance an immune response.

EXAMPLES

(228) The invention will now be further illustrated with reference to the following examples. It will be appreciated that what follows is by way of example only and that modifications to detail may be made while still falling within the scope of the invention.

Example 1—Construction of LAMP Constructs

(229) The improved LAMP Constructs illustrated in FIG. 1 can be constructed using standard molecular biology techniques well known to the skilled artisan. For example, plasmids comprising the polynucleotides can be designed to generate the different structures ILC-1 to ILC-6 shown in FIG. 1. The LAMP domains illustrated in FIG. 1 can be derived from the amino acid sequences shown in FIGS. 3-10. Preferably the LAMP domains are derived from the human LAMP proteins shown in FIGS. 3-10. The boundaries of each domain can be derived from FIG. 2A and FIG. 2B. It is envisioned that the corresponding domains can also be cloned from the orthologous sequences by identifying the equivalent domains when compared to the human sequence. The Allergen X (SEQ ID NO:Y) can be cloned into the described LAMP Constructs either individually or in combination as described in Table 1/FIG. 14.

Example 2—Immune Response Evaluation of Mice to LAMP Constructs

(230) The ability of the improved LAMP Constructs as described in Example 1 can be tested for their ability to modulate an immune response. For example, Female BALB/c mice can be immunized i.d with 50 ug of the improved LAMP Constructs in 100 ul PBS using nanopass on day 0, 7, and 14. Experiment will then be terminated 2 weeks after the last dose.

(231) Splenocytes (3×10^5 /well) are stimulated with allergenic protein (10 ug/ml) in T cell media (RPMI with 10% heat inactivated FBS, 1% penicillin/streptomycin, and 1 $\times$ 2-ME), supernatants are collected 48 h after. Supernatants are diluted (400 ul supernatant+200 ul T cell media) and cytokines are evaluated by ELISA. IL-10 or IL-4 production can be measured via ELISPOT assay.

(232) Alternatively, serum samples can be diluted 1:100 (day 21), 1:2000 (day 35) or 1:5000 (day 56) fold in 1% BSA in PBS. Day 56 samples are further diluted by a 7 point 1:3 serial dilution to measure the endpoint antibody titers. To detect IgE, sera can be treated with Agarose-Protein G (Thermo Fisher Scientific, Rockford, IL) 50 minutes and then 1:20 diluted samples are loaded to ELISA plates. Samples are detected with goat anti-mouse IgG1-HRP, goat anti-mouse IgG2a-HRP (Southern Biotech, Birmingham, Al), or rat anti-mouse-IgE-biotin (R35-118, BD Pharmingen, San Jose, CA) followed Pierce Streptavidin-HRP (Thermo Fisher Scientific, Rockford, IL). Reaction is developed with SureBlue TMB Substrate and stopped with TMB Stop Solution. Plates are read (OD450) by using Epoch ELISA reader (BioTek, Winooski, VT). Endpoint titers are determined by subtracting twice above the of background average (PBS) reading. The means and standard errors of endpoint titers or OD450 values per group are analyzed by using Excel statistic function. IgE data is analyzed by using Student T test. Tests are two tailed, and p values<0.05 were considered significant. After three doses of vaccination, antigen-specific IgG1 and IgG2a antibodies are expected to be observed in the recipient mice. It is believed that IgG2a levels will be much higher than expected due to the predicted Th2 to Th1 skewing of the immune response.

Example 3—Prime/Boost Protocol

(233) Herpesvirus entry mediator (HVEM), also known as tumor necrosis factor receptor superfamily member 14 (TNFRSF14) or CD270, is a human cell surface receptor of the TNF-receptor superfamily. In recent years, HVEM has been found highly expressed on hematopoietic cells and a variety of parenchymal cells, such as breast, melanoma, colorectal, and ovarian cancer cells, as well as gut epithelium. HVEM is a bidirectional protein, either inhibiting or stimulating T cells, through binding to BTLA or LIGHT (TNFSF14).

(234) We generated a DNA vaccine encoding HVEM-LAMP to generate an antibody which could block the inhibitory function of HVEM for tumor therapeutic applications. We hypothesized that LAMP will promote the antibody response by enhancing the affinity of HVEM specific antibodies and/or expanding the repertoire of B cell epitopes in the HVEM protein. In this study, we compared the immunogenicity of HVEM encoding plasmid with and without LAMP (SEQ ID NO: 158 and SEQ ID NO:159). Plasmids encoding HVEM-LAMP and HVEM and recombinant HVEM protein were designed as described herein.

(235) Goat anti-mouse IgG-HRP was purchased from Southern Biotechnologies (Birmingham, AL). SureBlue TMB microwell peroxidase substrate and TMB stop solution were purchased from KPL (Gaithersburg, MD). ELISPOT plates were ordered from EMD Millipore (Billerica, MA, Cat. No. MAIPS4510). IFN- γ antibody pair used in ELISPOT was purchased from BioLegend (San Diego, CA) and clones AN18 and R46A2 were used as coating and detection, respectively. Streptavidin-HRP and AEC substrate were purchased from BD Biosciences (San Jose, CA).

(236) Six to eight week old female Balb/c mice were purchased from Harlan Laboratories (Frederick, MA) and maintained at animal facility in Immunomic Therapeutics, Inc. (Rockville, MA). Mice (n=6) were treated with 10 μ g/dose of HVEM-LAMP, HVEM, or LAMP vector control by electroporation IM delivery at days 0, 7, and 14. On day 35, mice were boosted with 5 μ g HVEM protein in the presence of Alum by i.p. injection. On day 28 and 49, mice were bled and sera were isolated for antibody detection. Mice were sacrificed on day 56 and splenocytes were

tested for IFN- γ production by ELISPOT.

(237) ELISA procedure was followed by Su et al., *J of Immunol Res*; (10):1-15 (2016). Plates were coated with 5 μ g/ml HVEM protein. Data were analyzed by using Microsoft Excel and Prism 6 software.

(238) The primary aim of this study was to compare the antibody profiles between HVEM-LAMP and HVEM. On day 28, HVEM-LAMP vaccinated mice produced significant higher level of HVEM specific IgG antibody than that of the HVEM group (FIG. 11). After a protein boost, the HVEM specific antibody was increased about 1000-fold in HVEM immunized mice and the mean titer was changed from 100 to 108000. This result indicates that the immune memory was induced by the HVEM DNA plasmid. Although HVEM DNA alone only induced a minimal antibody response, protein boost rapidly recalled the immune memory. On the other hand, HVEM-LAMP group again exhibited a significant higher titer than the HVEM and LAMP groups, the mean titer is 5 folds of the HVEM group, indicating the power of LAMP in enhancing antibody response (FIG. 12).

(239) Additionally, serum samples (Day 49) from HVEM+LAMP or HVEM alone immunized/HVEM protein boosted mice were pooled and tested for peptide mapping. Twelve peptides were found to be bound to the pooled serum (mouse IgG reaction) and seven of the twelve peptides showed strong binding affinity. HVEM+LAMP alters the binding affinity of peptides 17, 24, 25, and 28 as compared to HVEM alone as shown in FIG. 13. These changes may have physiological effects in protecting tumor growth.

(240) This example illustrates the ability to use a prime-boost protocol with the improved LAMP constructs described herein along with the described allergens.

Example 4—Testing of Amb a 1-hLAMP

(241) Ragweed, a major allergy risk factor, is one of the most important pollen allergens in North America and parts of Europe. Epidemiological studies showed that 23% to 32.8% of the US population is sensitized to ragweed, whereas prevalence of sensitization in European countries varies between 3.5% (eg, Italy) and 54% (eg, Hungary). Amb a 1, the most abundant allergen in ragweed pollen, is composed of a mixture of five isoforms with amino acid sequence identities ranging between 63% and 87%. The clinical relevance of Amb a 1 has been documented in numerous publications. DNA vaccination has great potential as an effective prophylactic and therapeutic solution to ragweed pollen allergy. We have previously integrated the advantages of the DNA vaccine technique with the MHC II pathway targeting property of LAMP-1 and designed a novel DNA vaccine against ragweed allergy.

(242) Here, we optimized our LAMP platform by replacing the hinge region of LAMP (ILC-4). This study aims to compare the in vivo immunogenicity of different version of Amb a 1-LAMP vaccines using the sequence of SEQ ID NO: 137 as the allergen. Control vector, Amb a 1-hLAMP (complete LAMP), Amb a1-hLAMP preluminal (ILC-1), Amb a1-hLAMP Hinge (ILC-4), and Amb a 1 protein were made by NTC (Lincoln, NE). Goat anti-mouse IgG2a-HRP and goat anti-mouse IgG1-HRP were purchased from Southern Biotechnologies (Birmingham, AL). SureBlue TMB microwell peroxidase substrate and TMB stop solution were purchased from KPL (Gaithersburg, MD). Mouse monoclonal anti-hLAMP was purchased from Origene Technologies (Rockville, MD). Rabbit monoclonal anti-GAPDH antibody was bought from Abcam (Cambridge, MA). Goat-anti-mouse and goat anti-rabbit secondary antibody were obtained from Sino Biological (Wayne, PA)

(243) Vaccines, adjuvants and immunizations. 40 μ g of Control vector, complete LAMP, ILC-1, and ILC-4 were used in a total volume of 20 μ l per mouse per dose for intradermal/electroporation injection. Mice were immunized with the vaccine by i.d. delivery on days 0, 7, and 14. Mice were bled on days 28 and day 40 for serum collection. Serum was collected and stored in -30° C.

(244) Western Blot. 293T cells were transfected with the plasmids using lipofectamine 2000 reagents (Invitrogen). Transfected cells were washed with PBS and suspended in 200 μ l of RIPA

lysis buffer with halt proteinase inhibitors (Thermo Scientific, Waltham, MA). Lysates were centrifuges (700 g for 15 minutes at 4° C.), followed by measurement of protein concentration in the clarified supernatants using Pierce BCA protein Assay kit (ThermoFisher Scientific, Waltham, MA). 5 µg of protein was electrophoresed in pre-cast (4-20%) SDS-PAGE gels (BioRad, Hercules, California), and transferred onto nitrocellulose membranes (BioRad).

(245) Membranes were blocked with Detection™ block buffer (KPL) and probed anti-human LAMP (FIG. 15A) or anti-GAPDH and goat anti-mouse-HRP or goat anti-rabbit antibody (FIG. 15B), followed by developing with TMB (KPL).

(246) Measurement of serum Amb a 1-specific IgG1 and IgG2a by ELISA. As shown in FIG. 16, the murine antibody response to Amb a 1 was assessed by indirect ELISA. ELISA plates (MaxiSorp) were coated with 5 µg/ml Amb a 1 in PBS buffer overnight and then blocked with 2% BSA in PBS. Plasma samples were diluted 1:300 or 1:1000 in blocking buffer. Samples were detected with goat anti-mouse IgG1-HRP or IgG2a-HRP. Reaction was developed with SureBlue TMB Substrate and stopped with TMB Stop Solution from KPL (Gaithersburg, MD). Plates were read (OD450) by using Epoch ELISA reader (BioTek, Winooski, VT).

(247) Statistics. Two-Way ANOVA test was performed using GraphPad Prism 6.0 software to evaluate the statistical significance. Data represent mean of antibody titers±SEM (n=9). Two-way ANOVA was used for statistical analysis. * p<0.05; ** p<0.01, *** p<0.001 **** p<0.0001.

(248) Results. In this study, we tested different constructs of Amb a 1-LAMP vaccines. After 3 dose of DNA vaccines (one week apart), we found ILC-1 and ILC-4 induced unexpectedly higher Amb a 1 specific IgG2a response at day 40 than the complete-LAMP Amb a 1 vaccine.

Example 5—Testing of Bet v 1-hLAMP

(249) Bet v 1, the major birch pollen allergen, is considered the prototype for the PR-10 protein family causing respiratory allergy. The greatest majority of birch allergy patients (over >90%) react to Bet v 1, and as a consequence is used as a marker for birch pollen allergy. DNA vaccination has great potential as an effective prophylactic and therapeutic solution to birch pollen allergy in early spring.

(250) Here, we tested protein expressed from control vector Bet v 1-hLAMP (complete LAMP), Bet v 1-hLAMP preluminal (ILC-1), Bet v 1-hLAMP Hinge (ILC-4), and Bet v 1 protein as described in Example 4. In this Example, the sequence of SEQ ID NO: 141 is the allergen. FIG. 15A and FIG. 15B demonstrate expression. As shown in FIG. 17, after 3 dose of DNA vaccines (one week apart), we found ILC-4 induced unexpectedly significantly stronger Bet v 1 specific IgG2a response at day 28 and day 41 than the complete-LAMP Bet v 1 and ILC-1 vaccine, suggesting this new version of Bet v 1-LAMP construct is very immunogenic.

Example 6—Testing of Fel d 4-hLAMP

(251) Cats are popular household pets and commonly cause allergies. Cat allergy is unique among allergy to mammals in that the major allergen Fel d 1 is a uteroglobin-like protein and not a lipocalin. However, Fel d 4, which has been identified as a lipocalin allergen produced by the cat, binds IgE at relatively high frequency in cat-sensitive individuals. The biochemical spectrum of the cat allergens is thus uncertain, particularly with regard to the role that a cat lipocalin protein may play in sensitization to cats in allergic individuals. Recently, Fel d 1- and Fel d 4-specific IgE was assessed in patients with pet allergy. Of those with cat allergy, 94% had increased levels (>0.35 kU/L) of Fel d 1, and 49% had increased levels of Fel d 4.

(252) Here, we tested protein expressed from control vector Fel d 4-hLAMP (complete LAMP), Fel d 4-hLAMP preluminal (ILC-1), Fel d 4-hLAMP Hinge (ILC-4), and Fel d 4 protein as described in Example 4.

(253) FIG. 15A and FIG. 15B demonstrate expression. In this Example, the sequence of SEQ ID NO:182 is the allergen. As shown in FIG. 18, after 3 dose of DNA vaccines (one week apart), we found ILC-1 induced slightly higher Fel d 4 specific IgG2a response at day 28 and day 41 than the other groups. However, there is no statistical difference between ILC-1 and ILC-4.

Example 7—Testing of Cry J 1

(254) The Japanese red cedar tree is a cultural symbol in Japan, but its pollen is a national scourge. An estimated 25% of the Japanese population, more than 25 million individuals, are allergic to Japanese red cedar pollen.

(255) We tested Cry J1 and Cry J2 specific antibody responses by ELISA. FIG. 15C. In this Example, the sequence of SEQ ID NO:224 is the allergen. Female BALB/c mice were immunized ID with 40 µg of control vector, CryJ1+CryJ2+Complete LAMP, or Cry J1+J2+ILC-4 DNA in 20 ul PBS via the ear on day 0, 7, and 14. The study was terminated 26 days after the last dose. Serum samples were collected at day 28 and day 40. Cry J1 and Cry J2 specific IgG1 and IgG2a were measured by indirect ELISA. Data represent mean of antibody titers±SEM. N=6 per group. Two way ANOVA was used for statistical analysis. *p<0.05,** p<0.01.

(256) The complete and ILC-4 single Cry J1+J2+LAMP constructs were compared. Cry J1 and Cry J2 specific IgG1 and IgG2a responses are summarized in FIG. 19. Although the ILC-4 construct showed a trend of higher titers with Cry J1, these were not statistically significant. In the Cry J2 specific response, however, the titers were statically higher than the Complete LAMP construct.

(257) In conclusion, data from this study suggest that two constructs were expressed in vivo and LAMP significantly improved the humoral immune response.

(258) Variations, modifications, and other implementations of what is described herein will occur to those of ordinary skill in the art without departing from the spirit and scope of the invention and the claims. All of the patents, patent applications, international applications, and references identified are expressly incorporated herein by reference in their entireties.

Claims

1. A polynucleotide encoding a lysosomal associated membrane protein (LAMP) Construct, wherein the LAMP Construct comprises: 1) two homology domains of a luminal domain of a LAMP protein, and 2) at least one allergen comprising CryJ1 and/or CryJ2, wherein the allergen is placed between the two homology domains.
2. The polynucleotide of claim 1, wherein the LAMP protein is selected from LAMP-1, LAMP2, LAMP-3, LIMP 2, Macrosialin, Endolyn, LAMP5 or LIMBIC.
3. The polynucleotide of claim 2, wherein the LAMP Protein comprises an amino acid sequence selected from any one of SEQ ID NO:1-113.
4. The polynucleotide of claim 2, wherein the LAMP Protein comprises an amino acid sequence at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, 96%, 97%, 98% or 99% identical to the amino acid sequence of any one of SEQ ID NO:1-113.
5. The polynucleotide of claim 1, wherein each allergen is separated by a linker.
6. The polynucleotide of claim 5, wherein the linker is selected from the amino acid sequence GPGPG (SEQ ID NO: 251) or PMGLP (SEQ ID NO: 252).
7. The polynucleotide of claim 1, wherein the LAMP Construct further comprises a Transmembrane Domain and/or a Cytoplasmic Tail of a LAMP Protein.
8. The polynucleotide of claim 1, wherein the LAMP Construct further comprises a signal sequence.
9. The polynucleotide of claim 8, wherein the signal sequence is derived from a LAMP Protein.
10. A host cell comprising the polynucleotide of claim 1 or comprising a vector comprising the polynucleotide.
11. A vector comprising the polynucleotide of claim 1.
12. The polynucleotide of claim 1, wherein the LAMP Protein is human LAMP-1.
13. The polynucleotide of claim 12, wherein the two homology domains comprise human LAMP-1 Homology Domain 1 and human LAMP-1 Homology Domain 2.

14. The polynucleotide of claim 13, wherein the human LAMP-1 Homology Domain 1 comprises: (a) the amino acid sequence of residues 29 to 194 of SEQ ID NO: 1, or (b) a variant of (a) wherein said variant comprises an amino acid sequence at least 95% or at least 97% identical to the amino acid sequence of (a); and/or wherein the human LAMP-1 Homology Domain 2 comprises the amino acid sequence of residues 228 to 381 or residues 228 to 382 of SEQ ID NO: 1.
 15. The polynucleotide of claim 7, wherein the Transmembrane Domain comprises residues 383-405 of SEQ ID NO: 1 and/or wherein the Cytoplasmic Tail comprises residues 406 to 417 of SEQ ID NO: 1.
 16. The polynucleotide of claim 1, wherein the allergen comprises: (a) the amino acid sequence of residues 22-374 of SEQ ID NO: 222, residues 23-514 of SEQ ID NO: 223, and/or SEQ ID NO: 224; or (b) an amino acid sequence that is at least 95% identical to the amino acid sequence of residues 22-374 of SEQ ID NO: 222, residues 23-514 of SEQ ID NO: 223, or SEQ ID NO: 224.
 17. The polynucleotide of claim 1, wherein the polynucleotide is DNA.
 18. The polynucleotide of claim 1, wherein the polynucleotide is RNA.
 19. The polynucleotide of claim 1, wherein the polynucleotide is a viral vector.
 20. The polynucleotide of claim 4, wherein the polynucleotide is a self-replicating RNA viral vector.
 21. An antigen presenting cell comprising the polynucleotide of claim 1.
 22. The antigen presenting cell of claim 21, wherein the cell is a dendritic cell.
 23. A pharmaceutical composition comprising (a) the polynucleotide of claim 1, (b) a vector comprising the polynucleotide, or (c) an antigen presenting cell comprising the polynucleotide.
 24. The polynucleotide of claim 1, wherein the allergen comprises an amino acid sequence comprising residues 22-374 of SEQ ID NO: 222 or an amino acid sequence at least 95% identical to residues 22-374 of SEQ ID NO: 222, and comprising residues 23-514 of SEQ ID NO: 223 or an amino acid sequence at least 95% identical to the amino acid sequence of residues 23-514 of SEQ ID NO: 223.
 25. The polynucleotide of claim 1, wherein the allergen comprises the amino acid sequence of SEQ ID NO: 224 or an amino acid sequence at least 95% identical to the amino acid sequence of SEQ ID NO: 224.
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